KENYA METHODIST UNIVERSITY

Distance Learning Material

AGEC 111

INTRODUCTION TO AGRICULTURAL ECONOMICS

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Contents

Teaching and learning methods

The course will be implemented using lectures, problem based learning and group discussions.

Students' assessment

The student will be evaluated as follows;

Assignment and cats-30%

Main Exam-70%

Assignment

1. In its agricultural recovery strategy for wealth and employment creation, the government of Kenya identified Agriculture as an important sector for realization of its objective of employment creation and poverty reduction. Explain the importance of agriculture from this perspective.
2. State and explain the reform strategies the government of Kenya intends to implement in order to revitalize Agriculture, as stipulated in the strategy for revitalizing agriculture(SRA) document, 2004-2015
3. Free markets are notoriously prone to price changes. Stabilization is therefore a common reason for government intervention in agricultural markets of developing countries.
   a) Using appropriate illustrations, explain how buffer schemes work.
   b) Outline the problems that buffer stock schemes face, and how such problems can be alleviated.
4. How do export subsidies differ from tariffs in the regulation of international trade? Use appropriate illustrations
5. Write short notes on the following:
   a) Lorenz curves as used in the illustration of income distribution
   b) Profit maximization

Introduction

As you begin your studies, you are probably wondering, why study economics? In fact, people do it for a number of reasons.

Many study economies because they will be considered illiterate if they cannot understand the laws of supply and demand.

Others are interested in learning about how computers and the information revolution are shaping out society or why inequality in the distribution of income in the United States has risen so sharply in recent years.

All these reasons and many more, make good sense. Still as we may come to realize, there is one overriding reason for learning economics. All your life from cradle to grave and beyond you will run up against the brutal truths of economics. As a voter, you will make decisions on issues that cannot be understood until you have mastered the rudiments of this subject. Without studying economics, you cannot be fully informed about international trade, the economic impact of the internet, or the trade-off between inflation and unemployment.

Choosing your lifes occupation is the most important economic decision you will make. Your future depends not only on your own abilities but also on how economic forces beyond your control affect your wages. Also, your knowledge of economics may help you invest the next egg you save from your earnings of course; studying economics cannot make you a genious. But without economics the dice of lie are craded against you.

There is no need to be labour the point. We hope you will find that in addition to being useful economics is a fascinating field in its own right. Generations of students, often to their surprise, have discovered how stimulating economics can be.

There is no single definition of economics which entirely covers all its diverse aspects but as a working definition, 'Economics can be considered as a social science which studies the allocation of scarce recourses that have potentially alternative uses among the competing and virtually limitless wants of consumers in society'.

Economists see all economic questions as deriving from scarcity and they study the ways in which people can cope with scarcity.

Agricultural Economics on the other hand is an applied social science that deals with the production, distribution and consumption of Agricultural or farm goods and services. It is a social science because it deals with people in their dairy activities where choices are required. In the overall economist is concerned with overcoming the effect of scarcity by improving the efficiency with which scarce resources are allocated among with other competing uses so as to best satisfy human wants.

Application of Economic theory to agricultural problems went through a process of slow acceptance in the world. To date the field is fully developed.

Micro economics and macroeconomics

Adam Smith is usually considered the founder of the files of microeconomics, the branch of economics which today is concerned with the behaviour of individual entities such as markets, firms and households. In the Wealth of Nations (1776), Smith considered how individual prices of land, labour and capital, and inquired into the strength and weaknesses of the market mechanism. Most important, he identified the remarkable efficiency properties of markets and saw that economic benefit comes from the self-interested actions of individuals. These remain important issues today, and while the study of micro economics has surely advanced greatly since Smith's day, he is still cited by politicians and economists alike.

The other major branch of our subject is macroeconomic, which is concerned with the overall performance of the economy. Macroeconomics did not even exist in its modern form until 1936, when John Maynard Keynes published his revolutionary General Theory of Employment, interest and money. A that time, England and the United States were still stuck in the Great Depression of the 1930's with over one- quarter of the American labor force unemployed. In his new theory Keynes developed an analysis of what causes business cycles with alternating spells of high unemployment and high inflation. Today, macroeconomics examines a wide variety of areas, such as how total investment and consumption are determined, how central banks manage money and interest rates, what cause international financial crises, and why some nations grow rapidly while others stagnate. Although macroeconomics has progressed far since his first insights, the issues addressed by Keynes still define the study of macroeconomics today.

The two branches- microeconomics and macro economics- converge to form the core of modern economics.

The three problems of economic organization

Every human society- whether it is an advanced industrial nation a centrally planned economy, or an isolated tribal nation – must confront and resolve three fundamental economic problems. Every society must have a way of determining what commodities are produced, how these goods are made, and for whom they are produced.
Indeed, these three fundamental questions of economic organization – what, how, and for whom are as crucial today as they were at the dawn of human civilization. Let's look more closely at them:

- What commodities are produced and in what quantities? A society must determine how much of each of the many possible goods and services it will make and when they will be produced. Will we produce pizzas or shirts today? A few high quality shirts or many cheap shirts? Will we use scarce resources to produce many consumption goods (like pizzas)? Or will we produce fewer consumption goods and more investment goods (like Pizza – making machines), which will boost production and consumption tomorrow?
- How are goods produced? A society must determine who will do the production, with what resources, and what production techniques they will use. Who farms and who teaches? Is electricity generated from oil, from coal, or from the sun? will factories be run by people or robots?
- For whom are goods produced? Who gets to eat the fruit of economic activity? Is the distribution of income and wealth fair and equitable? How is the national product divided among different households? Are many people poor and a few rich? Do high incomes go to teachers or athletes or autoworkers or venture

capitalists? Will society provide minimal consumption to the poor, or must people work if they are to eat?

Positive economics versus normative economics

In thinking about economic questions, we must distinguish questions of fact from questions of fairness. Positive economic describes the facts of an economy, while normative economic involves value judgments.

Positive economics deals with questions such as : why do doctors earn more than janitors? Does free trade raise or lower the wages of most American? What is the impact of computers on productivity? Although these are difficult question to answer, they call all be resolved by reference to analysis and empirical evidence. That puts them in the realm of positive economics.

Normative economics involves ethical precepts and norms of fairness. Should poor people be required to work if they are to get government assistance? Should unemployment be raised to ensure that price inflation does not become too rapid? Should be United States break up Microsoft because it has violated the antirust laws? There are no rights or wrong answers to these questions because they involved ethics and values rather than facts. They can be resolved only by political debate and decision. Not by economic analysis alone.

Major types of economic systems

Free market system

A free market system refers to an economic system whereby there is an absence of government intervention and where the forces of demand and supply are allowed to operate freely. The free play of market forces where the state plays little or no role in economic activity is the fundamental characteristic of a free market system.

Characteristics of a free market system

The following are the key characteristics of a free market system.

i) **The ownership of private property.** The institution of private property implies that individuals and organizations have the right to own, control and dispose of factors of production.

ii) Free market economies are characterized by **freedom of choice and enterprise**. Freedom of enterprise implies that individuals are free to buy and hire economic resources to organize these resources for production to sell their products in markets of their choice. Freedom of choice. Freedom of choice signifies that the owners of factors of production may use these resources as they prefer.

iii) **Self – interests**. This entails that each economic agent attempts to do what is best for itself. Firms will usually aim to maximize profits. Owners of factors of production will aim at obtaining the highest possible rewards and workers will tend to move to occupations that offer the highest remuneration. Consumes will attempt to maximize their utilities.

iv) **Competition**. The theoretical model of a free market economy is based on a situation whereby there are large number of buyers and sellers, implying that the market determines prices. Theoretically competition constitutes the regulatory mechanism of capitalism.

v) **The price mechanism**. Is the fundamental feature of a free market system and will therefore be explored in a little more depth.

The price mechanism

The allocation of resources via the price mechanism in a free market system can be analyzed in terms of the three fundamental choices facing any society.

i) **What to produce?** With respect to the problem of what to produce in free market system, the price mechanism only facilitates the production of those commodities for which consumers are wiling to pay a unit price sufficiently high to cover at least the full cost of producing them in the long run. It implies that by paying a higher price, consumers can usually induce producers to increase the quantity supplied of a given commodity. On the other hand, a reduction in price normally gives rise to a decrease in the quantity supplied.

ii) **How to produce?** In a free market economy, the price of a factor of production represents its relative scarcity, which suggests that the best technique of producing a good or service is the one that minimizes costs of production. It therefore follows that if the price of a factor of production rises in relation to price of other factors used in the production process, producers will switch to a production technique that uses less of the expensive factor in order to minimize their costs of production.

iii) **For whom to produce?** Regarding the question of whom to produce under the price mechanism, the economy will produce those commodities that satisfy the wants of those people who have money to pay for them. This suggests that the higher the income of the individual, the more the economy will be geared to producing the commodities of such a consumer.

Advantages of a free market system

i) A free market system is in line with concept of consumer sovereignty. According to the notion of consumer sovereignty, the consumer is considered the best judge of his or her own welfare. Producers undertake production in line with consumer preferences since it is in their own interests to do so. Consumers can, therefore, influence what goods and services are produced directly by their purchases.

ii) A free market system is characterized by a greater responsiveness to changes in the international economic environment. This is because profit – maximizing firms are exposed to much greater competition in international markets and hence they have an incentive to respond faster to changes in the relative prices of goods and services and factors inputs. For example, firms operating in market – directed economies were able to adjust faster and more effectively to the adverse impact of Organization of Petroleum Exporting Countries (OPEC) price hikes in the 1970s

iii) In a free market system, firms have a greater incentive to bear risks since profit incentives exist. This tends to encourage hard work and initiative, contributing to technological innovation. Technical progress in turn contributes to economic growth.

iv) A free market economy may encourage foreign investments because overseas producers could expect a higher return on their investment and are less concerned about government controls. A lower level of political risks is usually associated with economies that are market – oriented.

v) The efficiency of firms may increase since those firms that do not produce what consumers want at a low cost may go out of business. Firms are therefore encouraged to utilize resources efficiently.

vi) In a free market system, consumers have a greater choice of producers. They therefore have a larger number of goods and services to choose from. Increased competition arising from the existence of a number of producers may also contribute to an improved quality of commodities.

Disadvantages of a free market system

i) A free market system tends to give rise to the development of monopolies. A monopoly refers to a market situation where there is only one supplier. Monopolies tend arise because modern technology has made it possible for large scale producers to obtain many economies of scale. Private monopolies can potentially exploit the consumer through high prices and low quality.

ii) In a free market economy externalities are not taken into account. **Externalities** refers to social costs and benefits which are not fully accounted for in an market system. In a free market system the focus is on private costs and benefits. This gives rise to social costs such as pollution and noise which lead to reductions in welfare of people experiencing these costs by say living near a factory.

iii) A free market economy may not produce public goods. Public goods are those commodities characterized by **non – excludability** and **non – rivalry** in consumption. Non excludability implies that it is not possible to exclude individuals who have not paid from consuming the commodities. Non – rivalry in consumption implies that one person's consumption does not reduce the consumption of another individual. Examples of public goods are street lighting and defense. The problem of the provision of public goods by the private sector arises because of the **free rider problem.** The free rider problem leads to potential customers understating their preferences or willingness to pay and still gaining since they receive the good without necessarily paying for it. Such individuals gamble on the good being provided to others who will express some willingness to pay.

iv) A free market economy systematically under provides for merit goods like health and education. **Merit goods** are those commodities which are socially desirable. In a system where the price mechanism allocates resources merit goods will be provided inadequate quantities and would be too expensive for the vast majority of the population.

v) It is argued that in a free market economy certain socially undesirable goods and services known as **demerit goods** such as drugs and alcohol may be produced. This because the demand for such commodities exists and they may profitable to provide.

vi) A free market economy is likely to generate considerable inequalities in income and wealth. This is because where the price mechanism operates in factor markets the wages and salaries earned will depend on the forces of demand and supply. Those whose skills are in demand will therefore command much higher remuneration than those whose services are in low demand.

vii) It is argued that free market system subjects an economy to cyclical unemployment when production and consumption decisions get out of line. Labour market imperfections may also result in a form of unemployment known as structural unemployment.

Centrally planned economies

A **planned economy** is one where the crucial economic resource allocation decisions are determined by the state through an economic planning body which implements society's major economy goals. Although planned economies can take a variety of forms, their most important features include the following.

i) The allocation of resources is achieved by the use of am overall plan which sets production targets for different sectors of the economy that determine the supply of different commodities for a specified period of time. This master plan breaks down individual organizations.

ii) The rationing of certain commodities to predetermine the demand for them. Rationing refers to any method of allocating a scarce commodity other than by means of the price mechanism.

iii) The fixing of prices and wages by state bodies which are usually not equilibrium rates. These may, for example, take the form of maximum price controls and minimum wages.

iv) Often, but not always, economic resources are owned by the state. Factors of production such as land usually belong to the state and not to individuals.

v) The occasional existence of a conscripted labour market in which workers take jobs assigned to them.

The **advantages** of a planned economic system are as follows:-

i) All essential goods and services such as education and health are provided by the state regardless of whether consumers can afford to pay for them. There is therefore a lower possibility of them being under – provided in a planned economic system than in a free market system.

ii) The notion's wealth tends to be more evenly distributed than in a free market system. Since the means of production are publicly owned, there is little scope for individuals to acquire large amounts to wealth. In case a few do so, a system of public finance redistributes their wealth to the lower income groups.

iii) In a planned economy the state provides a check on monopoly power since private monopolies are not allowed to develop and those industries which can best be organized on a monopoly basis such as electricity and railways are controlled in the form of public monopolies.

iv) A planned economy takes account of the external costs and benefits of all activities. For example, an industry may not be established in a certain area on the basis that the industry imposes heavy social costs on the region in terms of industrial wastes and noise.

v) Since the prices of many goods and services are fixed, a much lower rate of inflation can be maintained than under the price mechanism where the prices as determined by the forces of demand and supply may be high.

vi) The economic fluctuations prevalent in free market economies may be reduced or stabilized by government macroeconomic policy. Thus, for example, in times of heavy unemployment the government may increase its expenditure on development projects, especially in affected regions.

The **disadvantages** of a planned economic system are as follows:-

i) There is a greater likelihood of resources being wasted in a planned economy than in a market economy since it is difficult for the state to assess consumer demand in advance without resorting to the price mechanism. Co-ordination problems results in frequent over and underproduction with its associated wastage.

ii) The cost of gathering information on what, how and for whom to produce is likely to be high requiring the expertise of professionals like statisticians, economist, engineers, planners and administrators. The costs of administering a system of planning may be such as to outweigh its benefits. This is a disadvantage relative to a free market economy where the price mechanism provides a costless source of information.

iii) In the absence of the profit motive prevalent in planned economies, there is no incentive for innovation and hard work which gives rise to insufficiency in terms of lower output per worker. Such reduced productivity may hamper economic growth.

iv) In command or planned economies, there is a limitation on consumer sovereignty since demand is manipulated to match the limited range of goods produced tend to be standardized with little regard for individuals tastes.

v) Even if the pattern of required information about allocation decisions is collected, the pattern of consumers' preferences and the society's composition of resources may well change before the production and distribution plans are implemented. There is time lag between the collection of information and the formulation of production plans based upon that information. There is further time lag between the implementation of production plans and the realization of production targets.

vi) The absence of competition also reduces efficiently since any loss which might arise from production of planned goods is borne by the state.

Mixed economic system

A mixed economic system is one that combines the features of a free market system with some degree of central control. In a mixed market economy the allocation of resources between alternate uses is determined largely by the individual actions through the price mechanism, but the authorities play some role in determining the aggregate level of output through the use of fiscal and monetary policies, and in influencing the distribution of income through progressive taxation and welfare legislation. In some cases, the government may take control of sectors of the economy by nationalizing certain industries. Such a blend of private and public control characterizes most of the contemporary economies, although private enterprise is heavily circumscribed in communist countries such as Cuba. A mixed economic system combines the advantages of a free market and of planned economic system while aiming to eliminate their disadvantages.

All real – life economies are mixed in that government planning and the price mechanism share the function of dealing with the fundamental economic questions. The United States economy, for example, is regarded as a mixed economy in which resources are largely privately owned and in which price mechanism is the fundamental means by which resources are allocated. The economies of Kenya, Uganda and Tanzania in the east Africa region are also examples of mixed economic systems. By contrast the Soviet Union until the considerable changes of the recent years was regarded as an example of a planned economic system. Only small proportion of resources allocation was allowed to be influenced by the price mechanism.

The particular mix of planning and reliance on the price mechanism in any economy is substantially influenced by the political ideology of the government in office at a given point in time.

Whatever mix a country decides upon, governments can influence the allocation of resources in the following ways:-

i) Through the use of a system of **taxes and subsidies.** A government can, for example, raise an indirect tax with the aim of shifting the use of resources from a particular line of production such as cigarette production. A subsidy, on the other hand, can be used by government to encourage a partivular line of production
ii) The government can influence resources allocation by the way in which **it spends income raised in taxes.** In many countries it is common, for example, for military equipment, schools and hospitals to be built by private firms under government contract.
iii) The government can influence the allocation of resources by **producing goods and services itself.** This is especially so in the case of public goods like defense and to some extent in the case of merit goods like health and education.
iv) The government can affect resources allocation through **income distribution.** In this case the government may levy higher taxes on the wealth and transfer the money to the less well – off sections of society.
v) The government can influence the resources allocation process through the use of **maximum and minimum prices.** These are alternatively referred to as the price ceiling and price floors respectively.

The concept of consumer sovereignty and its limitations

The concept of consumer sovereignty refers to the freedom of individuals and households to decide for themselves what they want to buy. Underlying consumer sovereignty is the idea that the consumer is the best judge of his or her own welfare. The preceding idea underlies most of consumer behavior and through it the greater part of economic analysis including the pareto optimum which is the most widely accepted optimum in welfare economics. Consumer sovereignty is limited by the following factors:-

i) **The nature of the economic system.** In general, the consumer is more sovereign in a free market oriented system where commodities are produced more in line with consumer preferences. In a planned economic system less regard is given to consumer preferences.

ii) **The size of the consumer's income.** In general. The larger the consumer's income, the greater is the consumer's sovereignty since the consumer can afford to choose from a wider range of goods and services which he or she afford to buy. Consumer sovereignty is, however, always limited to some extent by income since wants are limited.

iii) **The range of goods available.** Different consumers have very different individual tastes and preferences and it is difficult for the available range of goods and services to satisfy all the consumers.

iv) **Government intervention in providing merit goods.** The fact that the government need to intervene to provide essential goods and services demonstrates that the complete reliance on consumer preferences especially in a market – oriented economy would lad to the under provision of certain essential goods and services.

v) **The power of advertising.** Most notably advocated by Harvard professor Prof. J..K. Galbraith. Galbraith has argued that advertising by large corporations not only entices consumers to use the products of these corporations but also creates new wants.

vi) **The existence of monopoly.** A monopoly is said to exist where there is a single supplier of a given commodity. Monopolies often limit consumer sovereignty by providing relatively higher priced and lower quality commodities.

vii) **Consumer habits.** Individual consumers have different habits and are often reluctant to change these habits. Thus, for example, individual consumers often get attached to particular suppliers and particular products and are reluctant to change.

viii) **Provision of standardized goods.** It is often cheaper for manufacturers to produce standardized commodities, but in producing standardized commodities consumer freedom is inevitably limited since there is less regard for individual tastes and preferences.

ix) **Fashion and customs.** Consumer behavior is inextricably linked to the prevailing or latest trends in society and the reluctance to contravene established conventions limits consumer freedom to some extent.

What are the different ways that a society can answer the questions of what, how and for whom? Different societies are organized through alternative economic systems, and economics studies the various mechanisms that society can use to allocate its scarce resources.

We generally distinguish two fundamentally different was f organizing an economy. At alone extreme, government makes most economic decision, with those on top of the hierarch giving economic commands to those further down the ladder. At the other extreme, decisions are made in markets where individuals or enterprises voluntarily agree to exchange goods and services, usually through payments of money. Let’s briefly examine each of these two forms of economic organization.

In the United States, and increasingly around the world, most economic questions are settled by the market mechanism. Hence their economic systems are called market economies. A market economy is one in which individuals and private firms make the major decisions about production and consumption. A system of prices, of markets, of profits and losses, of incentives and rewards determines what, how, and for whom. Firms as produce the commodities that yield the highest profits (the what) by the techniques of production that are least costly (the how). Consumption is determined by individuals’ decisions about how to spend the wages and property incomes generated by their labor and property ownership (the for whom). The extreme case of a market economy, in which the government keeps its hands off economic decisions, is called a laissez- faire economy.

By contrast, a command economy is one in which the government makes all important decisions about production and distribution. In a command economy, such as the one which operated in the soviet Union during most of the twentieth century, the government owns most other means of production (land and capital); is also owns and directs the operations of enterprises in most industries; it is the employer of most workers and tells them how to do their jobs; and it decides how the output of the society is to be divided among different goods and services. In short, in a command economy, the government answers the major economic questions through its ownership of resources and its power to enforce decisions.

No contemporary society falls completely into either of these polar categories. Rather, all societies are mixed economies, with elements of market and command. There has never been a 100 percent market economy (although nineteenth-century England came close. Today most decision in the United States are made in the market place. But he government plays an important role in overseeing the functioning of the market; governments pass laws that regulate economic life, produce educational and police services and control pollution. Most societies today operate mixed economies.

DEMAND, SUPPLY AND EQUILIBRIUM

Basic elements of supply and demand

Like the weather, markets are dynamic, subjects to periods of storm and calm, and constantly evolving. Yet also with weather forecasting, a careful study of markets will reveal certain forces underlying the apparently random movements. To forecast prices and outputs in individual markets, you must first minister the analyses of supply and demand.

The theory of supply and demand shows how consumer preferences determine consumer demand for commodities, while business costs are the foundation of the supply of commodities.

This lesson introduces the notions of supply and demand and shows how they operate in competitive markets for individual commodities. We begin with demand curves and then dissents supply curves. Using these basic tools we will see how the market price is determined where these two curves intersect- where the forces of demand and supply are just in balance. It is the movement of prices - the price mechanism - which brings supply and demand into balance or the customers through the substitution effect.

Economics is the study of how societies use scarce resources to produce valuable commodities and distribute them among different people.

Behind this definition are two key ideas in economics: that goods are scarce and that society must use its resources efficiently. Indeed, economics is an important subject because of the fact of scarcity and the desire for efficiency.

Consider a world without scarcity. If infinite quantities of every good could be produced or if human desires were fully satisfied, what would be the consequences? People would not worry about stretching out their limited incomes because they could have everything they wanted; business would not need to fret over the cost of labor or health care; Government would not need to struggle over taxed or spending or pollution because nobody would care. Moreover, since all of us could have as much as we pleased, no one would be concerned about the distribution of incomes among different people or classes. In such an Eden of affluence, all goods would be free, like sand in the desert or seawater at the beach. All prices would be zero, and markets would be unnecessary. Indeed, economics would no longer be a useful subject.

But no society has reached a utopia of limitless possibilities. Ours is a world of scarcity, full of economic goods. A situation of scarcity is one in which goods are limited relative to desires. An objective observer would have to agree that, even after two centuries of rapid economic growth, production in the United States is simply not high enough to meet everyone's desires. If you add up all the wants, you quickly find that there are simply not enough to meet everyone's desires. If you add up all the wants, you quickly find that there are simply not enough goods and services to satisfy even a small fraction of everyone's consumption desires. Our national output would have to be many items arger before the average American could live at the level of the average doctor or major-league baseball player. Moreover, outside the United States, particularly in Africa, hundreds of million s of people suffer from hunger and material deprivation.

Given unlimited wants, it is important that an economy make the best use of its limited resources. That brings us to the critical notion of efficiency.

Efficiency denotes the most effective use of a society's resources in satisfying people's wants and needs. By contrast, consider an economy with unchecked monopolies or unhealthy pollution or government corruption. Such an economy may produce a distorted bundle of goods that leaves consumers worse off than they otherwise could be – either situation is an inefficient allocation of resources.

In economics, we say that an economy is producing efficiently when it cannot make anyone economically better off without making someone else worse off.

The essence of economics is to acknowledge the reality of scarcity and then figure out how to organize society in a w ay which produces the most efficient use of resources. This is where economics makes its unique contribution.

The theory of demand and supply enables us to understand the determination of prices and quantities in different markets. For example, why do the prices of agricultural commodities such as tomatoes, apples, mangoes and cabbages increase and decrease at certain times of the year? Why have the prices of computers, music systems and television sets been steadily declining over time? An understanding of the workings of the price system provides us with the answers to some of these questions. The price with the basis for determining the prices of both commodities like food and cars and also the basis for determining the prices of factors of production.

Definition of demand

Demand refers to the quantity of a commodity that consumers are willing and able to purchase at any given price over a given period of time. It is important to realize that demand is not the same thing as want, need or desire. Only when want is supported by the ability and willingness to pay the price does it become an effective demand and have an influence on the market price. Hence demand economics means effective demand.

The law of negatively sloped demand

This law states that "*Ceteris paribus*" the lower the price of a commodity, the greater the quantity demanded by the individual and vice versa".

Exceptions to the law of demand

There are some demand curves that slope upwards from left to right showing that as prices of a product rise more is demanded and vice versa. These type of demand curves are known as **regressive exceptional** or **abnormal** demand curves and occur in the following situations:-

i) When there is **fear of a more drastic price change in the future.** This will cause consumers to increase their quantity demand to avoid paying an even higher price in the future. This situation is often found in the stock exchange where there is often an increase in the demand for the shares of a company if its share price is expected to increase.

ii) In **case of a giffen goods.** these refer to basic foodstuffs that constitute a high proportion of the budget of low income families. When the price of giffen good rises, the proportion of the total income of individuals who consume these giffen goods rises and since such consumers are worse off in real terms, they can no longer afford to consume other more expensive commodities like meat and fruits. To make up for the foods they can no longer afford to buy, they are likely to purchase more of basic foodstuffs. Conversely, when the price of basic foodstuffs falls, they become better off in real terms and are likely to buy more of relatively more expensive foodstuffs and less of basic foodstuffs.

iii) **Goods of ostentation** (**veblen goods**). These are commodities whose prices fall in the upper price ranges and that have a snob appeal. The wealthy are usually especially concerned about status, believing that only goods at high prices are worth buying and have the effect of distinguishing them from other consumers. In the case of such commodities, a firm increasing its prices may find that sales of its product increase and at a lower prices less of the commodity may be bought as the commodity is rejected as being sub – standard. Consumers often in making comparisons between two similar products with different prices opt for the relatively more expensive one believing it to be better. Examples of goods of ostentation could be expensive perfumes and jewellery.

The determinants of demand

The demand for a product can be considered from the standpoint either individual demand or market demand. The determinants individual demands are as follows: -

i) **The price of the product.** When deciding whether or not to buy a particular product, an individual will compare the price of the product with the amount of utility or satisfaction that he or she expects to receive from the product. If the price is considered worth the anticipated utility, the individual will buy the product and if not, he will not buy it. A decrease in the price of a product will probably increase an individual's demand for it since the amount of utility obtained is more likely to be worth the lower price. Conversely, a rise in the price of a product will probably result in fall in demand as the amount of utility received is less likely to be worth the higher price to be paid. An example of this phenomenon is the hotel industry in Kenya. There is usually an increase in domestic tourism during the low season when many Kenyan consider the lower hotel prices to be worth the level of satisfaction they are receiving. During the high season when the hotel prices are high, many Kenyans do not consider the satisfaction they receive to be worth the higher prices and hence they do not travel.

   If the amount that consumer is willing and able to purchase changes due to a change in the price, a change in the "quantity demanded" is said to take place. If on the other hand the amount the consumer is willing and able to purchase changes because of a change in any of the determinants of demand other than price then a change in "demand" is said to take place. A more profound analysis into why a change in the price of a given commodity leads to a change in the quantity demanded will be undertaken later when utility analysis and indifference curve analysis is studied.

ii) **The price of related goods.** which may be either substitutes or complements. Two goods X and Y are said to be **substitutes** if a rise in the price of one commodity, say Y leads to rise in the demand of the other commodity X. Eg.
   A mango and an orange

Demand for some commodities can also be affected by changes in the prices of complementary goods. two goods A and B are said to be **complementary** if a rise in the price of one of the goods, say, A leads to a fall in the demand of another good, say B. Complementary goods are usually jointly demanded in the sense that the use of one requires or enhanced by the use of the other. Example of complementary goods is Hamburgers and chips

iii) Demand is influenced by changes in **disposal real income.** An individual's level of income can be said to have an important effect on his or her level of demand for most products. If income increases the demand for most goods and services will increase especially the demand for better quality goods and services. A rise in income may, however, cause the demand for some goods to fall. Such commodities are referred to as inferior goods and comprise items like basic foodstuffs.

iv) Demand is influenced by changes in **tastes and fashion.** Personal tastes play an important role in governing a consumer's demand with certain consumers, for example, preferring to consume imported commodities despite their being much more expensive than local commodities. Tastes may also change according to season of the year. Prevailing fashions are an important determinant of tastes. The demand for clothing, for example, particularly susceptible to changes in fashion.

v) The level of **advertising** is also an important determinant of demand. In highly competitive markets, a successful advertising campaign will increase the demand of a particular product while at the same time decreasing the demand for competing products.

vi) **The availability of credit to consumers.** The factor especially affects the demand for durable consumer goods which are often purchased on credit. Changes in terms on which credit can be obtained will have a marked effect on the demand for certain products like furniture and electrical appliances. For example, a decrease in the availability of credit or the introduction of more stringent credit terms is likely to lead to a reduction in the demand for some durable consumer goods.

vii) **Government policy.** The government may influence the demand of a given commodity through legislation.

The market demand for a commodity

The **market** or **aggregate** demand for a commodity provides the alternative amounts of the commodity demanded per time period, at various alternative prices, by **all** the individuals in the market.

Price elasticity of demand

Let's look first at the response of consumer demand to price changes:
The price elasticity of demand (sometimes simply called price elasticity) measures how much the quantity demanded of a good changes when it prices changes. The precise definition of price elasticity is the percentage change in quantity demanded divided by the percentage change in price.

Goods vary enormously in their price elasticity, or sensitivity to price changes. When the prices elasticity of a good is high, we say that the good has “elastic” demand, which means that its quantity demand responds greatly to price changes. When the price elasticity of a good is low, it is “inelastic” and its quantity demanded responds little to price changes.

For necessities like food, fuel, shoes, and prescription drugs demand tend to be inelastic. Such items are the staff of life and cannot easily be forgone when their prices rise. By contrast, you can easily substitute other goods when luxuries like European holidays, 17 – year-old scotch whiskey, and Italian designer clothing rise in price.

Goods that have ready substitutes tend to have more elastic demand than those that have no substitutes. If all food or footwear prices were to rise 20 percent tomorrow, you would hardly expect people to stop eating or to go around barefoot, so food and footwear demands are price-inelastic. On the other hand, if mad-cow disease drives up the rice of British beef, people can turn to beef from other countries or to lamb or poultry for their meat needs. Therefore, British beef shows high price elasticity.

The length of time that people have to respond to price changes also plays a role. A good example is to price changes also plays a role. A good example is that of gasoline. Suppose your are driving across the country when the price of gasoline suddenly increase. It is likely that you will sell your car and abandon your vacation? Not really. So in the short run, the demand for gasoline may be very inelastic.

In the long run, however, you can adjust your behaviour to the higher price of gasoline. You can buy a smaller and more fuel-efficient car, ride a bicycle. Take the train, move closer to work, or carpool with other people. The ability to adjust consumption patterns implies that demand elasticties are generally higher in the long run that in the short run.

Economic factors determine the size of price elasticities for individual goods: elasticities tend to be higher when the goods are luxuries when substitutes are available, and when consumers have more time to adjust their behavior.

Calculating elasticities.

If we can observe how much quantity demanded changes when price changes, we can calculate the elasticity. The precise definiantion of price elasticity, E_D is the percentage change in quantity demanded divided by the percentage change in quantity demanded divided by the percentage changed in price. For convenience, we drop the minus signs so elasticities are all positive. We can calculate the coefficient of price elasticity numerically according to the following formula:

Price elasticity of demand = E_D

$$= \frac{\text{percentage change in quantity demanded}}{\text{percentage change in price}}$$

Now we can be more precise about he different categories of price elasticity:

- When a 1 percent change in price calls for the more than a 1 percent change in quantity demanded, the good has price –elastic demand. For example, if a 1 percent decrease in quantity demanded, the commodity has a highly price-elastic demand.
- When a 1 percent change in price produces less than a 1 percent change in quantity demanded, the good has price-inelastic demand. This case occurs, for

instance, when a 1 percent increase in price yields only a 0.2 percent decrease in demand.

- One important special case is unit – elastic demand, which occurs when the percentage change in quantity is exactly the same as the percentage change in price. In this case, a 1 percent increase in price yields a 1 percent decrease in demand.

1.

Elasticity and revenue

Many businesses want to know whether raising prices will raise or lower revenues. This question is of strategic importance for businesses like airlines exporting horticultural products, baseball teams, and magazines, which must decide whether it is worthwhile to raise prices and whether the higher prices make up for lower demand. Let's look at the relationship between price elasticity and total revenue.

Total revenue is by definition equal to price times quantity (or P x Q). If consumers buy 5 units at $3 each, total revenue is $15. If you know the price elasticity of demand, you know what will happen to total revenue when price changes:

1. When demand is price-elastic, a price decrease reduces total revenue.
2. When demand is price-elastic, a price decrease increases total revenue.
3. In the borderline case of unit-elastic demand, a price decrease leads to no change in total revenue.

The concept of price elasticity is widely used today as businesses attempt to separate customers into groups with different elasticity's. This technique has been extensively pioneered by the airlines. Another example is software companies, which have a wide range of different prices for their products in an attempt to exploit different elasticity's. For example, if you are desperate about buying a new operating system immediately, your elasticity is low and the seller will profit from charging you a relatively high price. On the other hand, if you are not in a hurry for an upgrade, you can search around for the best price and your elasticity is high. In this case, the seller will try to find a way to make the sale by charging a relatively low price.

Definition of supply

Supply refers to the quantity of a given commodity that a producer is willing and able to sell at a given price over a specific time period. The supply schedule and supply curve demonstrates the relationships between market prices and the quantities that suppliers are prepared to offer for sale. Supply differs from "existing stock" or the "amount available" in that it is concerned with the amounts actually brought to the market. The basic law of supply states that "A greater quantity will be supplied at a higher price than at a lower price".

Determinants of supply

i) **Price of the good:** As the price of a given commodity say X rises, with all costs and prices of all other goods remaining unchanged, the production of commodity X becomes more profitable. The existing firms are therefore likely to expand their output while new firms are likely to be attracted into the industry. It should be noted, however, that not just current prices but prices but also expectations concerning future prices may motivate the procedures in questions.

ii) **Price of certain other goods:** Changes in the prices of other commodities may affect supply of a commodity whose price does change. If the price of another good increases because of, say, a rise in demand the production of the good whose price is relatively higher will become more profitable to produce and resources will tend to move towards its production. A producer of barley, for example, who sees that the price of wheat has risen may decide to use more of his land or wheat production. Barley and wheat are said to be substitutes in production. The extent to which firms can move from one industry to another in search of higher profits depends on the occupational and geographical mobility of the resources.

iii) **Prices of factors of production:** As the prices of factors of production used intensively by the producers of a certain commodity rise, so do the firm's costs. This will cause the supply to fall since some firms will reduce output while other less efficient firms will make losses and eventually leave the industry. Similarly, if the price of one factor of production, say land, should increase some firms may move out of the production of land – intensive in some other factor of production which is relatively cheaper.

iv) **The state of technology:** Technological improvements or progress such as improvements in machine performance, management and organization or an improvement in the quality of raw materials leads to lowering of costs through increased productivity and increases the profit margin on every unit sold. This chould lead to an increase in supply.

v) **Indirect taxes and subsidies:** A tax on a commodity can be regarded as an increase in the costs of supplying that commodity and the supply curve will shift to the left as a result. Subsidies usually take the form of payments by governments to producers and will have the effect of lowering the costs of production and hence increasing the supply.

vi) **Weather:** The supply of agricultural products is considerably affected by change in weather conditions. Output in agriculture is subject to variations from one year to the next, which are independent of the acreage planted. An excellent growing season associated with favorable weather conditions will result in a bumper harvest leading to an increase in supply. An unfavorable season that results in a poor harvest may be viewed as an increase in the average cost of production because a given expenditure on inputs yields a smaller output than it would in a good season. A bad harvest is represented by a leftward shift of the supply curve.

vii) **Incidence of strikes:** Strikes will lead to a reduction in the supply of a product. The supply of manufactured goods is particularly likely to be affected by a industrial disputes because of generally stronger unions in the industrial sector.

The market supply of a commodity

The **market** or **aggregates** supply of a commodity refers to the alternative amounts of the commodity supplied per period of time at various alternative prices by **all** the producers of this commodity in the market. The market or aggregate supply of a commodity depends on all the factors that determine the individual producer's supply and additionally on the number of producers of the commodity in the market.

Price elasticity of supply

Of course, consumption is not the only thing that changes when prices go up or down. Businesses also respond to price in their decisions about how much to produce. Economists define the price elasticity of supply as the responsiveness of the quantity supplied of a good to its market price.

More precisely, the price elasticity of supply is the percentage change in quantity supplied divided by the percentage change in price.

As with demand elasticity's, there are polar extremes of high and low elasticities of supply. Suppose the amount supplied is completely fixed, as in the case of perishable fish brought to market to be sold at whatever price they will fetch. This is the limiting case of zero elasticity, or completely inelastic supply, which is a vertical supply curve.

At the other extreme, say that a tiny cut in price will cause the amount supplied to fall to zero, while the slightest rise in price will coax out an indefinitely large supply. Here, the ration of the percentage change in quantity supplied to percentage change in price is extremely large and gives rise to a horizontal supply curve. This is the polar case of infinitely elastic supply.

Between these extremes, we call supply elastic or inelastic depending upon whether the percentage change in quantity is larger or smaller than the percentage change in price. In the borderline unit elastic case, where price elasticity of supply equals 1, the percentage increase of quantity supplied is exactly equal to the percentage increase in price.

You can readily see that the definitions of price elasticities of supply are exactly the same as those for price elasticities of demand. The only difference is that for supply the quantity response to price is positive. While for demand the response is negative. The exact definition of the price elasticity of supply,

$$E_s = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

The concept of equilibrium in economics

Equilibrium in economics refers to a situation in which the forces that determine the behaviour of some variable are in balance and therefore exert no pressure on that variable to change. In equilibrium the actions of all economic agents are mutually consistent. **Market equilibrium** occurs when the quantity of a commodity demanded in the market per unit of time equals the quantity of the commodity supplied to the market over the same period of time. Geometrically, equilibrium occurs at the intersection point of the commodity's market demand and market supply curve. The price and quantity at which equilibrium exists are known as the equilibrium price and equilibrium quantity respectively.

Maximum price controls

Maximum price controls are imposed below the equilibrium price since government considers that the price as determined by the forces of demand and supply is too high. The implications of maximum price controls can be illustrated in the diagram below:

[h1]

In the figure above, P_e and Q_e are the price and quantity as determined by market forces. The government considers that the maximum price P_e as determined by the market forces of demand and supply is too high, especially for low – income consumers. The government therefore decides to fix a disequilibrium price such as P_{MAX} such that the price has to be legally reduced from P_e to P_{MAX}. The maximum price can therefore, benefit low income consumers and can be used as one of several non – inflationary measures aimed at dealing with the negative effects of inflation.

If the government imposes a maximum price control given by P_{MAX} there will be shortage of the commodity given by $(Q_2 – Q_1)$. Some of the disequilibrium consequences are the following:

Consumers, especially the poorer members of the community will be helped by the lower price, but suppliers are making less of the commodity available. This will lead to queues and shops being more emptied more quickly.

A black market is likely to develop where the commodities are sold above the legal price. Suppliers may illegally be able to sell the quantity Q_1 for price P*. The attraction of such high illegal prices may lead to the production of unmonitored extra suppliers that are of inferior quality.

Consumer will not be able to secure all their wants at P_{MAX} and as result excess demand will spill over into other related market and prices will be pushed up in those markets.

Maximum price controls are also likely to give rise to demand for centrally administered system of rationing since producers would otherwise sell commodities of a “first come served basis of using sellers preferences”. However, a system of rationing is very expensive to implement.

Additionally, producers will move way from price – controlled lines of production. Research and development in line of production where prices controlled is likely to be discouraged. In the case of maximum price controls applied to essential commodities this may imply that production may shift to luxury goods where there are no price controls. Moreover, firms that cannot recover their costs from the revenue provided by the maximum price will leave price leave price – controlled industries with a consequent falls in employment.

Minimum price controls

Minimum price controls are imposed above the equilibrium price since the government considers that the price as determined by the forces of demand and supply is too low. Minimum price controls can be show in figure below

[h2]

In the figure above the minimum price as determined by the forces of demand and supply is P_e which is considered too low by the government which then fixes a minimum price P_{MIN} above the equilibrium price.

Minimum price controls will have the following **likely effects:** In the case of **minimum producer prices** there will be a stabilizing effect on producers incomes. The minimum price is designed to ensure that prices for certain products do not fall when market forces change to create unfavorable conditions for the product.

Minimum price controls in the case of minimum producer prices will lead to excess supply or a glut of the commodity as given by ($Q_4 – Q_3$) in figure 2.20. If the product is not perishable it may be stored for release during less favorable harvests. If the product is perishable, then it may have to be disposed of by the authorities at price below the minimum price such as P*.

An additional example of minimum price controls is **minimum wage legislation** which aims at ensuring that workers earn a reasonable income.

Thirdly, price controls may contribute to industrial peace especially if they constitute part of a comprehensive incomes' policy. This is because the control of wage increases may be accompanied by the controls of the price of certain basic commodities.

Fourthly, price controls may in certain cases be associated with a decrease in price and increase in output. This, for example, would be the case of a monopolist who is charging an exploitive price and is forced to lower his price which he accompanies by an increase in output, to compensate for the loss of revenue occasioned by the price decrease.

The effects of price decontrol

Price decontrol or **price liberalization** refers to a transition from a system whereby prices are fixed by the government to one where prices are determined by the market forces of demand and supply. In many countries, including Kenya, there has been a tendency to decontrol the prices of commodities whose prices were previously controlled as part of the general liberalization process of the economy. There effects of price decontrol are as follows:-

i) Prices in decontrolled industries tend to rise at first. This is because in the case of maximum price controls those were below the equilibrium prices. Consumers would have to pay higher prices for the products of most decontrolled industries as the prices tend towards the equilibrium levels.

ii) Decontrol of price may encourage greater investment by producers in decontrolled lines of production since they can charge higher and more profitable prices.

iii) It may encourage research and development since producers can now raise prices to recover their investment in this area. This could in turn foster economic growth by encouraging innovation.

iv) It is likely to lead to a more efficient long run allocation of resources in the case of both maximum and minimum price decontrols since there would be an absence of excess demand.

v) In the long – term as more firms enter decontrolled lines of production this is likely to lead to lower consumer prices because of increased competition.

vi) Higher prices imply an increased tax capacity as producers earn a higher income. This factor contribute to an increase in government revenue.

vii) The decontrol of minimum producer prices will lead to an increased fluctuation of producer prices especially in the agricultural sector with consequent fluctuations in producer incomes.

viii) The decontrol of minimum wages may lead to increase in employment

THE THEORY OF PRODUCTION

An introduction

Consumer and producer goods and services

Production comprised all those activities that provide the goods and services which people want and for which they are prepared to pay a price. The composition of total output can be classified into consumer goods and services.

Consumer goods

Consumer goods are commodities that satisfy out wants directly, which implies that we want them for their own sake. They can be considered to be of two types:

i) **Durable consumer goods:** These produce a steady stream o satisfaction while their values diminish relatively slowly through age and use, for example, furniture, books and domestic electrical appliances.

ii) **Non-durable consumer goods**: These are consumed or destroyed in the very act of being used, for example, food and cigarettes.

Producer goods

These are commodities that do not directly satisfy wants since they are not wanted for their own sake but for the contribution they make to the production of other goods. Examples of product goods are factories, buildings and cranes.

Services

Services are intangible economic goods such as transport, tourism, banking, insurance and administration. In developed economies a large proportion of the output consists of services. A distinguishing characteristic of services is that they are non-transferable, which implies that they cannot be purchased and then resold at a different price.

It is possible to subdivide production into three categories:

i) **Extractive industries:** Examples of such industries are farming, mining, fishing and forestry. Primary products result from such industries.

ii) **Manufacturing industries:** These include engineering, vehicle manufacture, chemicals and food processing.

iii) **Distribution industries:** These incorporate the activities of wholesaling and retailing.

Factors of production

Factors of production refer to inputs or resources of society used in the process of production. Classifications of factors of production differ, but in general, the following classification is recognized:

Land

Land is taken to refer to all the **natural resources** over which people have the power of disposal and which may be used to yield an income. Land includes farming land, building land, forests, rivers, lakes and mineral deposits. The total supply of land in the world is limited, although the supply of land for some particular use is not fixed. Thus, for example, more maize can be planted at the expense of potatoes. Alternatively, more land can be allocated to buildings at the expense of farming land, drainage, irrigation and fertilizers can increase the area of cultivated land.

Labor
Labor refers to the exercise of human **mental** and **physical** effort in the production of goods and services. It should be noted that labor is distinct in that I is the services of labor that are bought and sold, not the labor itself. Labor is also unique in that it is the reason why economic activity takes place. The supply of labor in an economy is a measure of the number of hours of work which is offered at given wage rates over a given period of time.

Capital
Capital is a man-made input. It can be classified as **working capital** or **circulating capital** referring to stocks of raw materials, partly finished goods and finished goods held by producers. Alternatively, capital can be classified as **fixed capital** which consists of equipment used in production such as replacement of worn-out producer goods every year is referred to as **depreciation**. The total output of producer goods is referred to as **gross investment** and any addition to capital stock is referred to as **net investment**. This implies that: Gross investment – Depreciation = Net investment

Note: In economics, the concept of depreciation is distinct from the concept of depreciation in accounting. In economics, depreciation refers to the annual replacement investment, whereas in accounting depreciation is considered to be the periodic allocation of the cost of an asset to expense even though this is dome partly to show the reduction in the value of a fixed asset.

Entrepreneurship
Entrepreneurship refers to the **organization** of all the factors of production with a view to profit. The entrepreneur fulfils two vital functions:

Firstly, the **hiring** and **combining** of the other factors of production, including making the decisions relating to what to produce, how to produce and where to produce.

Secondly, the **risk-taking** function which arises because most production is undertaken in anticipation of demand where the future is uncertain. Entrepreneurs make payments to cover their costs without any certainty that the costs will be covered by revenue.

Specialization

Specialization refers to the concentration of activity in those lines production where the **individual**, **firm** or **country** has some natural or acquired advantage. Adam Smith drew attention to the importance of division of labor in his book The Wealth of Nations. He was fundamentally concerned with division of labor of a particular industry where the manufacture of products was broken down into many specialized activities and Adam Smith observed that the making of pins required 18 distinct operations; estimating that the production per day in this factory was about 5,000 pins per person employed. If, however, the whole operation was undertaken from start to finish by each employee, smith estimated that he would have been able to make only a few dozen pins per day. Apart from specialization in particular industries the following other forms of specialization can be identified:

i) **International specialization** which refers to the concentration by a country of its resources on a specific area of production such as, for example, the concentration by Kenya in the production of coffee and tea or the production of copper by Zambia.

ii) **Regional specialisation** within a country where factor endowments and economic history have led industries to concentrate in certain areas because it is difficult for competitive plants to be established elsewhere. Thus, for example, in Kenya the production of tea is concentrated in the highlands.

iii) Specialisation between **industries** since each economy includes many industries each producing different commodities. For example, it is possible to speak of the motor manufacturing industry, the electrical component industry, the steel industry and so on.

iv) The specialisation between **firms** since industries are composed of firms that can be regarded as units of production. Thus, for example, different firms can specialize in the manufacture of different components of a product. Thus, for example, in the car industry certain firms specialize in the production of tyres while other firms specialise in the provision of spare parts.

v) The specialisation within factories which arises because one firm will often control many **factories,** and these are usually referred to as plants and are units of production. For example, motor manufacturer might find it economical to build engines in one plant, axles in another, car bodies in a third and so on, and subsequently transport all the parts to another plant for final assembly.

vi) Specialisation within plants can be of **two** types:
   a) A particular plant may produce more than one item and the plant may thus be regarded as two working side by side.
   b) Within every plant there is considerable specialisation of labour. In a typical manufacturing plant some employees will be receiving and storing raw materials and components whereas the majority will be engaged in the manufacturing process, while others will check and pack the finished product.

The advantages of specialization

i) The fundamental advantage of division of labour is the increased output arising from division of labour. With division of labour, a given set of factors of production can produce a far greater volume of goods and services than is possible without it. In the example given by Adam Smith, a small factory employing 10 men could produce perhaps 48,000 pins per day by dividing the work between them, whereas each might have difficulty in producing 20 pins a day if he had to do the entire task. It would not be an exaggeration to say that without industrial division of labour economic growth would be minimal.

The remaining advantages of the specialization of labour can easily be considered as factors contributing to the vast increase in output associated with modern productive methods.

ii) Since each person is endowed with different aptitudes and skills, an economic system that breaks down work into a variety of jobs gives individual workers the opportunity to exploit their particular talents fully. With specialisation people are more likely to find their optimal occupations.

iii) Specialisation improves the skill of workers given that one a particular person establishes himself in the job he usually finds that his skill on the job increases with practice. Constant repetition leads to greater dexterity.

iv) No time is wasted moving from one job to another in that the necessity of putting down one set of tools and picking up another is eliminated. Time saved by not moving between jobs implies that more will be produced.

v) There is a saving of time in the training of operatives since individuals can be trained more quickly for the performance of a single task than for the performance of multiple tasks. This saving of time is translated into a higher level of output.

vi) Specialisation makes possible the greater use of machinery since the breaking up of a complex process into a series of separate and simpler tasks makes it easier to design a machine to carry out the individual operations.

vii) Specialisation reduces cost per unit given that not all costs of production rise as output rises. Specifically fixed costs do not rise over large ranges of output and unit costs are, therefore, reduced.

viii) Specialisation increases the leisure time available since it allows same quantity of goods to be produced for a lower expenditure or resources. Specialisation, therefore, enables workers to enjoy more leisure time as well as more goods.

ix) Specialisation enables labour to be released to other industries because the greater productivity enables some units of labour to be released in order to develop greater skills. This would not be possible if each everyone aimed to be self-sufficient.

Disadvantages of specialisation

i) Specialisation involves close interdependence given that no one can specialise in making part of a product such as a car unless others specialise in making the other parts of the product. Additionally, people involved in production in one sector do so on the assumption that there are people involved in the production of other commodities in other sectors of the economy. This interdependence can be disadvantageous in that when disruptions our in one part of plant other parts are affected since operatives cannot pass on the goods they finished before the breakdown while those further down the production line find they are receiving no raw materials.

ii) Specialisation may lead to boredom or monotony as some workers perform the same operation hundreds of times. This monotonous repetition may lead to a greater incidence of accidents and greater absenteeism associated with a lower level of motivation. In addition, labour relations between senior management and production workers may deteriorate as communication becomes more difficult.

iii) Specialisation may be accompanied by a decline in craftsmanship as skills are transferred from the hands of the worker to a machine that controls the design, quantity and quality of a given product. It should, however, be noted that mechanization has produced many new craftsmen in occupations that require high degree of skill.

iv) Specialisation is associated with an increased risk of unemployment as specialised workers do not have a wide industrial training that would make them adaptable to changes in techniques o production. In the short-term such workers are susceptible to unemployment, although in the long-term the simplification of tasks implied by specialisation would mean that jobs indifferent industries are fairly similar and retraining is relatively easily achieved.

v) Specialisation is economically limited by the extent of the market in that methods of production using expensive capital equipment are only worthwhile f there is a potential demand for the mass produced product that keeps the capital equipment fully employed. This mass market may not always exist because of low income levels in a particular country.

vi) Specialisation especially in cases involving mass production inevitably associated with standardization of products. This is because of the heavy development costs incurred in launching mass produced commodities that leave little possibility for the accommodation of tastes and preferences of individual consumers. Consumer sovereignty is, to some extent, limited by this narrow regard for individual preferences.

Mobility of factors or production

There are two aspects of mobility of factors of production:

i) **Occupational mobility** which refers to the ease of movement of factors of production from one job task to another.

ii) **Geographical mobility** which relates to ease of movement of factors of production from one location to another.

Mobility of factors of production can be analyzed in terms of the individual factors or production.

Land

Land is not mobile in the geographical sense but has a high degree of occupational mobility. Land is occupational mobile in that it can be put to different uses such as putting up buildings, farming, roads and railways. However, some land such as land in arid areas has very limited occupational mobility.

Capital

Some capital is mobile in both senses, thus, for example, vehicles and tools are occupationally and geographically mobile. Some other categories of capital are immobile in both senses. Thus, for example, railways and blast furnaces are immobile both geographically and occupationally. Still other forms of capital may exist that are mobile in the occupational sense but immobile in the geographical sense, for example, buildings.

Labour

Labour is theoretically mobile in the geographical and occupational sense. In practice, however, many barriers to occupational and geographical mobility of labour exist.

Barriers to mobility of labour

The following are the barriers to geographical mobility of labour:

i) The cost of movement such as removal costs and the costs of buying and selling a house which can often be sustained.

ii) Housing shortages especially of rented accommodation in urban areas.

iii) The reluctance to break existing social ties and having to face the prospect of establishing new social relationships.

iv) The reluctance of families to move at crucial stages in their children's schooling especially if different parts of a country are operating under different systems of education.

v) Adverse geographical and climatic conditions that may not favour the health of workers.

vi) Language barriers both at local and international levels may be a hindrance to mobility. Different parts of a given country may have different local dialects. The official languages in various countries may also differ.

vii) Insecurity and political instability in certain areas may hinder mobility.

viii) An ignorance of job opportunities in different parts of the country maybe a factor contributing to reduced labour mobility.

The following are barriers to **occupational** mobility of labour:

i) The fact that different people possess different natural abilities and some people may not possess the attributes required by particular professions.

ii) Certain professions require long periods of education and training, for example, medicine and architecture and this may require a considerable financial sacrifice. In addition, the time period may be a deterrent in itself.

iii) Capital may be required to enter certain professions especially businesses where capital is needed to purchase of a practice or partnership requires capital.

iv) Class may be a barrier to occupational mobility n that a particular social background with education at certain schools may provide advantages in certain fields or employment.

v) Some trade unions or professional associations have established regulations that restrict entry into their professions, for example, law, medicine, engineering, architecture and building economics.

vi) Not all workers are knowledgeable about job opportunities especially where there is no organizational system to provide information on the existing opportunities.

Policies to assist mobility

The following policies can assist geographical mobility of labour:

i) The provision of information on regional job opportunities. This can be done effectively through the establishment of employment exchanges and job centres that would enable employers and potential workers to be more fully informed.

ii) Employers could enhance geographical mobility by assistance wit the cost of movement, for example, by meeting some of the removal costs.

iii) Employers can provide a hardship allowance particularly if movement to remote areas is involved.

iv) Geographical mobility can be enhanced by linking movement to promotion in the sense that a higher remuneration is offered to workers whom employers want to persuade to move to other areas.

v) Geographical mobility can be enhanced by employers providing low-cot housing in areas where they want employees to relocate.

The following policies can enhance **occupational** mobility of labour:

i) The provision of retraining schemes or centres where people bearing the burdens of economic change through unemployment can learn new skills which they can use to find alternative employment.

ii) The arrangement of financial assistance for people to enable them to start their own businesses. This can take the form of low-interest loans for projects that are considered to be viable.

iii) The provision of information on opportunities available in different occupations through, for example, employment exchanges. This policy is based on the idea that many workers are occupationally immobile because they are unaware of openings in other sectors of the economy for jobs which they are capable of doing with or without training.

iv) Legislation to reduce trade union and professional barriers to entry into different occupations. Such legislation would make entry requirements into different professions less stringent thereby requirements into different professions less stringent thereby giving entry access to a wider spectrum of people.

v) Industry can play an important role in reducing labour immobility by offering training to new entrants especially since current production technology in industry has given rise to more jobs that are of a semi-skilled nature and require only a few weeks of training.

Entrepreneurship

This is probably the most mobile factor of production in that the fundamental functions of the entrepreneur are common to all industries. Regardless of the type of economic activity there will always be a need to organize factors of production and take the associated risks or production. In order to search for profit opportunities, the entrepreneur will often be willing to move from one location to another and to change from one task to another.

The significance of mobility of factors of production

The mobility of factors of production is significant for the following reasons:

i) Mobility enables different factor combinations to be used. Thus, for example, more labour and capital can only be used if either of these factors is mobile to facilitate a change in the production technique. This enables producers to search for a least cost method of production.

ii) The mobility of factors of production facilitates the movement of factors of production from surplus to deficit areas. This signifies that if factors are sufficiently mobile, unemployment will be avoided in surplus areas whole production will be enhanced in deficit areas. This leads to a more efficient utilization of resources.

iii) The mobility of factors of production enables the benefits of economic growth to be spread more evenly throughout a country. Thus, for example, many industries are located in urban areas primarily because of the urban market and the advantages economies of scale. If industries can be encouraged to locate in rural areas through particular incentives then the benefits of industrials development in a particular country can be spread more evenly.

iv) Mobility of factors of production also enables the transfer of expertise to areas where it is deficient. Thus, or example, in the event of mobility experienced managers can contribute to the development of aspects of the firm lacking in managerial expertise and can even in some cases transfer their skills internationally. This is common in multinational corporations where managers can transfer their expertise from one subsidiary to another.

v) The possibility of vertical occupational mobility of labour can have motivational effects in that if workers perceive chances of being promoted for outstanding work, they are in general, likely to be much more efficient at their work thereby contributing to the overall efficiency of their business enterprise.

vi) Mobility maybe significant in that if workers are occasionally allowed to perform different tasks from the ones they usually perform and are capable of performing them, the they are less likely to experience the monotony often associated with specialisation, and its accompanying negative effects.

vii) Mobility is significant in the production of factors of production which are immobile occupationally, in that they have no alternative use, have no opportunity cost and are therefore not considered by the economist to be scarce resources.

viii) While a measure of mobility is essential, excessive mobility may be inefficient and costly. This is because each time a worker leaves his job and a replacement is required, the employer incurs certain costs in the form of a fall in output, costs of training the new worker and the loss of a skilled worker's output while the new entrant is being trained.

Combining factors of production

In making a product, a firm can usually vary the proportions in which it combines the inputs used to make the product. Thus, for example, many crops can be grown using capital-intensive or labour-intensive methods. It is important to study the effects of varying the proportions between factors because nearly all short run changes in production involve some changes in these proportions. Quite often, if a firm wants to change its output, it is unable to change the input of all factors or production. For example, a manufacturing firm wishing to increase its output is unable to build a bigger factory overnight.

The production horizon or span of production can be divided into the short and the long run.

In economics, the **short run** refers to a period of time in which only some variables change or economic processes work. It is a concept that can only strictly ve defined in the particular context in which it is applied since its meaning depends on which variables or processes the user of the term has in mind as being flexible. Its most common use is in the theory of the firm where it is defined as the period of time where the input of at least one factor of production cannot be varied and production has to be increased by using more of variable factors, say, labour, raw materials or fuel. The specific period of time considered as the short run varies with the application of the term given that, for example, it takes different mounts of time to build the plant and machinery for the different industries.

The **long run** is a period of time in which all variable are able to settle at their equilibrium or final disequilibrium levels and al economic processes have time to work in full. Its most common application is in the theory of the firm where it is considered to be the period of time in which the quantities of all factors of production employed can be varied and all entry and exit that can occur into or from an industry has occurred. The duration f the long run will depend on the context in which the term is applied, depending in the speed with which the variables spoken of change.

The concept of a production function

A production function is the technical relationship between the output of a good and the inputs required to make that food. The function may take the form of an equation, table or graph showing the maximum quantity of the commodity that can be produced per unit of time for each of a set of alternative inputs required to make that good.

A simple agricultural production function may be obtained by using various alternative quantities of labour per unit of time to cultivate a fixed amount (size) of land and recording the resulting alternative outputs of the commodity per unit of time. In this case, land is the fixed factor of production whereas labour is the variable factor of production. Since at least one factor of production is fixed the period in question is the short run.

Definition of key concepts

Average product is the output per unit of the variable factor.

$$\textbf{Average product} = \frac{\text{Total product}}{\text{no. of units of the variable factor (for example, workers)}}$$

$$AP = \frac{TP}{L}$$

Where labour (L) is taken to be the variable factor **Marginal product** refers to the changes in total product brought about by varying the employment of the variable factor by one unit for example, increasing employment by one person.

$$\textbf{Marginal product} = \frac{\text{Change in total product}}{\text{Change in quantity of labour employed}}$$

$$MP = \frac{\Delta TP}{\Delta L}$$

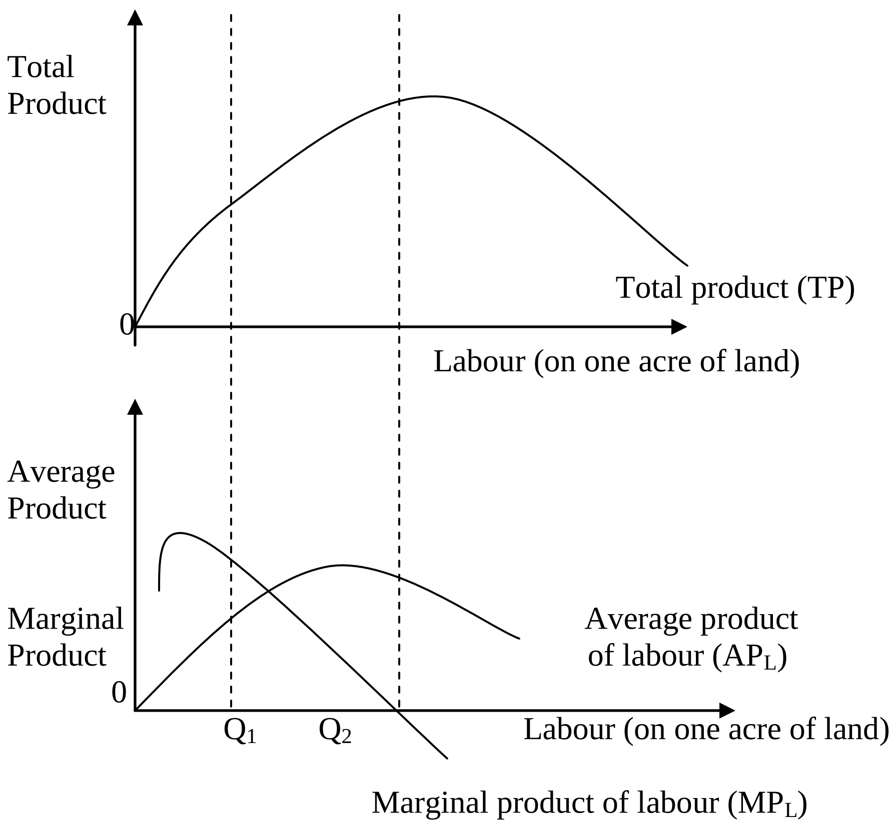
A hypothetical production function for small scale coffee farmer can be shown in the following table.

To illustrate a hypothetical production function for a small scale coffee farmer.

Land	Labour	Total Product (TP)	Average product of labour (AP_L)	Marginal product of labour (MP_L)
1	0	0	0	-
1	1	3	3	3
1	2	8	4	5
1	3	12	4	4
1	4	15	3.75	3
1	5	17	3.40	2
1	6	17	2.83	0
1	7	16	2.29	-1
1	8	13	1.625	-3

The shapes of the average and marginal product curves

The shapes of APL and MPL curves are determined by the shape of the corresponding TP curve which has the shapes shown below.



The relationship between the total product, average product and marginal product curve.

In the figure above the AP_L curve rises at first, reaches a maximum, then falls, but it also remains positive as long as the TP is positive. The MP_L curve also rises at first, reaches a maximum (before the AP_L reaches its maximum) and then declines.

The MP_L becomes zero when the TP is maximum and negative when the TP begins to decline. The falling portion of MP_L curve illustrates the law of diminishing returns.

The law of diminishing returns

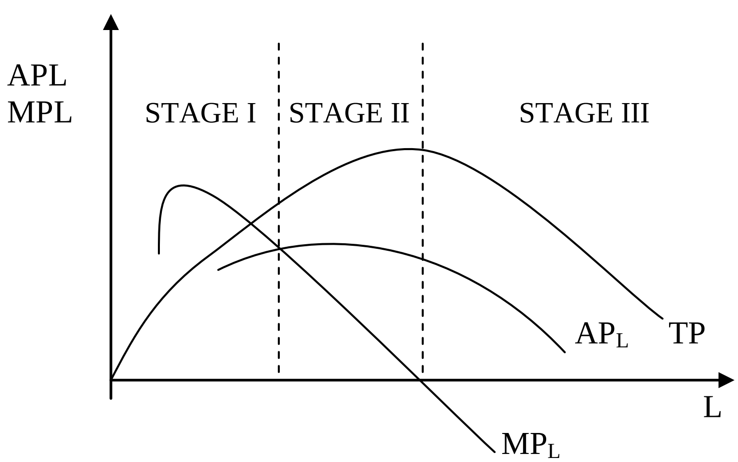
The law of diminishing returns states that " **Ceteris Paribus, as additional units of a variable factor are added to a given quantity of a fixed factor, total output will initially increase at an increasing rate, but beyond a certain level of output it will increase at a declining rate, and will eventually decline. The law of diminishing returns is also known as the law of variable proportions.**

Key assumptions of the law of diminishing returns

i) The state of technology is unchanged.
ii) Successive units of the variable factor are assumed to be equally efficient.
iii) Production takes place in the short run where at least one factor of production is fixed.
iv) There is one variable factor of production under consideration.

The stages of production

The stages of production implied by the law of diminishing returns can be illustrated in the diagram that follows:



The stages of production

The stages of production are as follows:

Stage 1

In stage 1 there are increasing returns to the variable factor. In this stage, total product is increasing at an increasing rate while marginal and average products are also rising with marginal product higher than the average product at any point. This is an indication of the increasing efficiency of the proportion in which the factors are combined since fixed factors are still underutilized and there is scope for greater specialisation.

Stage 2

Stage 2 represents decreasing returns to the variable factor in that total product is increasing at a decreasing rate. Marginal product and average product are positive but are falling during this stage, with the average product higher than the marginal product. Stage 2 goes from where the average product of labour is maximum to the point where the marginal product of labour is zero. Stage 2 is the only stage of production for the rational producer.

Stage 3

Stage 3 represents the stage of negative returns to the variable factor. At this stage marginal product is negative and as a result total product is declining. This represents a stage of extreme inefficiency when factors of production, are probably getting into each other's way. The producer will not operate in stage 3 even with free labour since he or she could increase total output by using less labour on one acre of land.

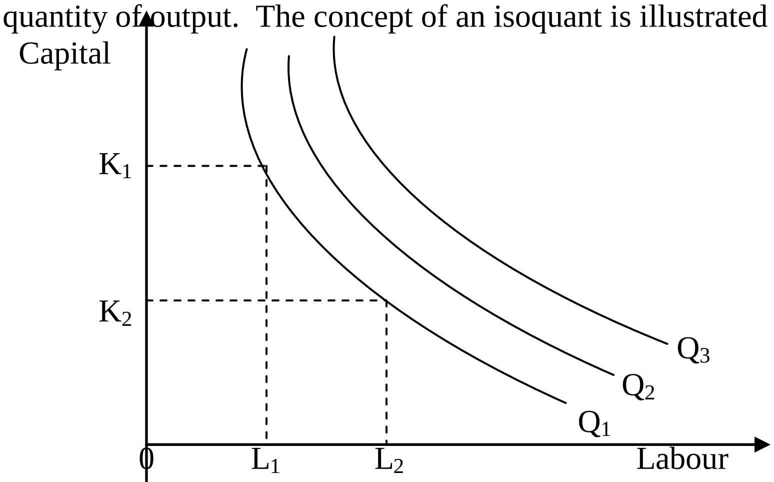
The least-cost factor combination

The law of diminishing returns, shows the tendency of output per unit of the variable factor when the proportions between the factors are varied. However, in order to determine the most profitable way of combining factors of production, one has to consider their prices as well ad their productivity. Physical productivity deals with technical efficiency. Entrepreneurs are more concerned with economic efficiency, and output and input will therefore be expressed in monetary terms. Assuming an objective of profit maximisation, the aim of entrepreneurs will be to maximize the difference between revenue and costs. Entrepreneurs will be interested in maximizing the productivity of a factor of production if that factor of production is expensive relative to other factors of production.

In order to analyse the least-cost combination we will introduce the concepts of an isoquant and an isocost line.

The concept of an isoquant

As isoquant shows al the different combinations of labour and capital with which a firm can produce a specific quantity of output. An isoquant is drawn on the assumption that there are only two factors of production, labour and capital. It is also assumed that it is possible to substitute capital for labour and vice versa continuously in the production process. A higher isoquant refers to greater level of output and a lower one to a smaller quantity of output. The concept of an isoquant is illustrated in the diagram below:



The concept of an isoquant

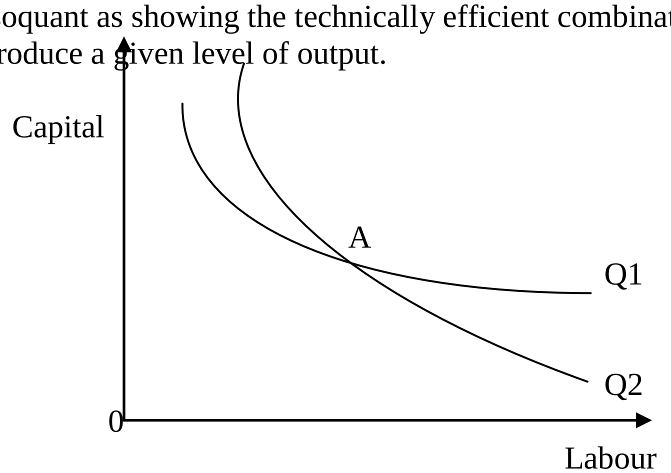
The figure above shows an isoquant map which refers to a family of isoquants that illustrates graphically the production function for a good. In the diagram, isoquants Q_2 and Q_3 represent higher levels of output than isoquant Q_1. Isoquants have three important properties: Firstly, no two isoquants interest; secondly, isoquants normally slope downwards from left to right; thirdly, they are convex to the origin.

The properties of isoquants

i) **Isoquants cannot intersect.**

It can be demonstrated that isoquants do not intersect by showing the absurd situation that would arise if they did.

Figure below shows two intersecting isoquants Q_1 and Q_2. Point A represents a combination of capital and labour which when used efficiently can produce two quantities of a given commodity Y. This result contradicts the very definition of an isoquant as showing the technically efficient combinations of capital and labour that produce a given level of output.



Two intersecting isoquants of commodity Y

ii) **Isoquants are negatively sloped**

This rises from the fact that both capital and labour are assumed to have positive marginal products which implies that the employment of extra units increases total output. It follows that to maintain a given level of output when the quantity of one factor is reduced, the quantity of the other must be increased. In figure above, if a lower quantity of capital K_2 is used to produce the level of output Q_1, then the quantity of labour employed must be raised form L_1 to L_2 to maintain the same level of output.

iii) **Isoquants are convex to the origin**

This property is related to the idea that labour and capital are not perfect substitutes for each other. this implies, for instance that as bigger quantities of capital are employed to produce a given level of output, labour becomes less and less capable of substituting for capital. In the same way, as greater quantities of capital and smaller quantities of labour are employed to produce a given level of output, capital becomes less and less capable of substituting for labour.

The marginal rate of technical substitution

The slope of an isoquant measures the rate at which labour can substitute for capital, keeping output constant. This slope is termed as marginal rate of technical substitution of capital for labour.

The marginal rate of technical substitution of labour for capital refers to the amount of capital that a firm can give up by increasing the amount of labour used by 1 unit and still remain on the same isoquant. The marginal rate of technical substitution of labour for capital is also equal to MP_L/MP_K. As the firm moves down an isoquant, the marginal rate of technical substitution of labour for capital diminishes. This is so because the less capital and the more labour the firm is using, (in other word, the lower point on the isoquant) the more difficult it becomes for the firm to substitute labour or capital in production.

This concept can be illustrated in the following table

To illustrate the marginal rate of technical substitution for a given isoquant.

L	K	$MRTS_{LK}$
1	10	-
2	7	3
3	5	2
4	3.5	1.5
5	2.5	1.0
6	1.8	0.7

The concept of an isocost line

As **isocost line** shows all the different combinations of labour and capital that affirm can purchase, given the total cost outlay of the firms and factor prices. This tool, together with the isoquant map, enables one to identify the cost minimizing combination of factors of production that a profit-maximising firm will employ to produce its chosen level of output.

Let C = cost outlay,
w = price of labour and
r = price of capital.

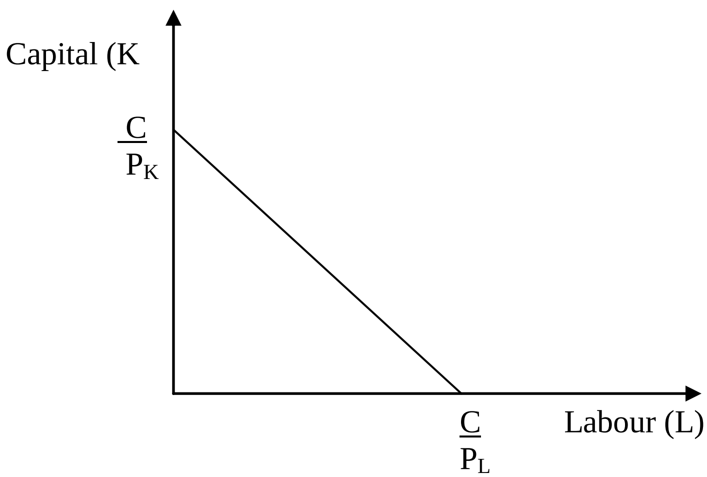
Such that $C = wL + rk$

$$Rk = C - wL$$

$$K = \frac{C}{r} - \frac{w}{r}L$$

$$\frac{\Delta K}{\Delta L} = \frac{-w}{r} = -\frac{P_L}{P_K}$$

An isocost line is illustrated below

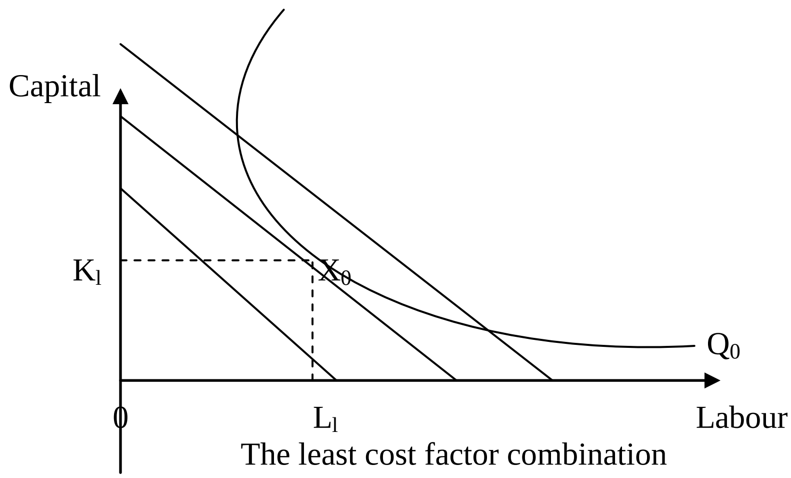


An isocost line

The slope of an isocost line is given by P_L/P_K, where PL refers to the price of labour and P_K to the rice of capital. If the firm spent all its total outlay on capital it would purchase C/P_K units of capital. If it spent all of its outlay on labour, it would purchase C/P_L units of labour. Joining these points by a straight line, we get the isocost of a firm. The firm can purchase any combination of labour and capital shown on its isocost.

The firm's least cost factor combination

A firm may wish to minimise the cost of producing a given level of output. It may, for example, want to produce Q tones of a commodity with a minimum cost outlay. The firm will minimize the cost of producing Q by employing the combination of capital and labour at which the isoquant is tangent to the lowest isocost line. This can be shown in the figure below.



The least cost factor combination

Points below X_0 are desirable because they show lower cost but are not sufficient for producing output Q. Points above X_0 could use higher cost to produce output Q. Hence X_0 is the least cost combination of capital and labour for producing Q_0.

Returns to scale

In the long run all the factors of production can be carried. This implies that a given level of output maybe achieved by varying all the inputs.

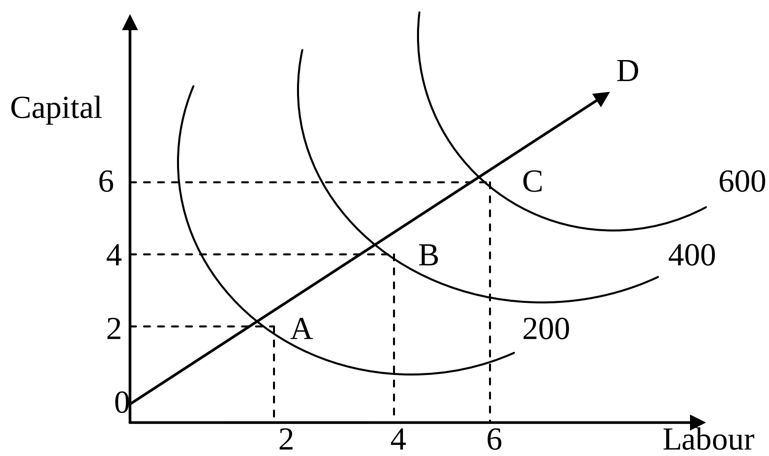
The concept of returns to scale refers to changes in output as all the factors are changed by the same proportion. It is a **long run** concept of production. Three fundamental concepts can be identified in this context.

i) Constant returns to scale. This arises when all inputs are increased in a given proportion. For example, if the quantities of labour and capital used per unit of time are both increased by 20 percent, output increases by 20 percent also. If labour and capital are doubled output also doubled. This concept can be illustrated using the numerical example below.

The concept of constant returns to scale

Labour (Number of people)	Capital (Number of machines)	Output (bags)
4	2	200
8	4	400

This concept can also be illustrated by isoquants in the figure below.



Constant returns to scale

The above figure shows returns to scale whereby if we double both inputs, we double output. Thus, OA = AB = BC. Output expands along the ray OD as long as relative factor prices remain unchanged.

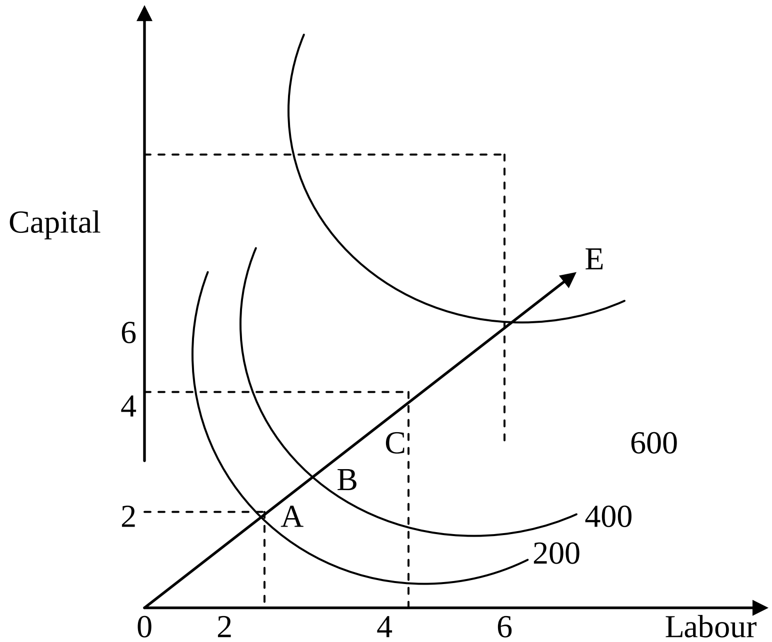
ii) **Increasing returns to scale**: This arises when all inputs are increased n a given proportion and the output increases more than proportionately. For example, if labour and capital are both increased by 20 percent, output rises by more than 20 percent. If labour and capital are doubles, output more than doubles.

This concept can be illustrated using the following numerical example:

The concept of increasing returns to scale

Labour (Number of people)	Capital (Number of machines)	Output (bags)
4	2	200
8	4	600

Increasing returns to scale may occur because increasing the scale of operations makes possible greater division of labour and specialisation. Greater specialisation may lead to an increase in labour productivity. In addition, a larger scale of operation may enable more specialized machinery to be used which was not feasible at a lower scale of operation. This concept can be illustrated by isoquants in the figure below.



Increasing returns to scale

Figure above shows increasing returns to scale whereby an increase in both inputs in a given proportion causes a more than proportionate increase in output. Thus,)A> AB>BC. If relative factors prices remain unchanged, output expands along the ray OE.

(iii). Decreasing returns to scale: This arises where all inputs are increased in a given proportion and output increase less than proportionately. For example, if labour and capital are both increased by 20 percent, output rises by less than 20 percent. This concept can be illustrated using the following numerical example:

Table 6.5 The concept of decreasing returns to scale

Labour (number of people)	Capital (Number of machines)	Output (bags)
4	2	200
8	4	300

Decreasing returns to scale may arise because as the scale of operations increases. Communication difficulties may arise which may complicate the effective running of the business. This concept can be illustrated by isoquants in figure below

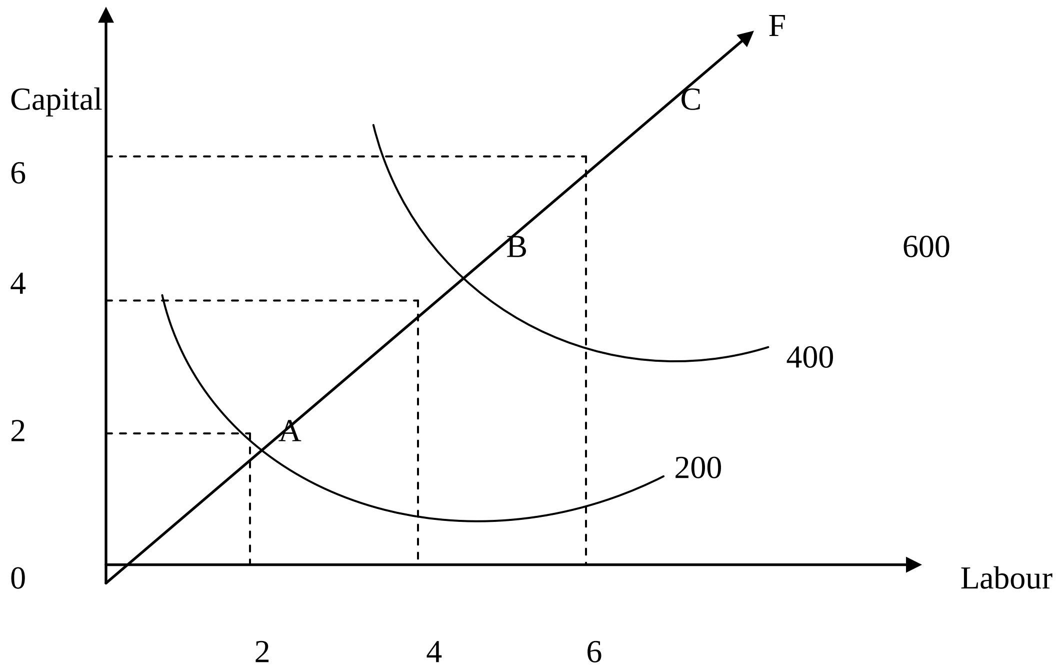


Figure above shows decreasing returns to scale. In this case, in order to double output, the firm must more than double the quantity of both inputs. Thus OA<AB<BC. It is likely that in expanding the scale of operations, a producer will at first experience increasing returns to scale as he or she is able to take advantage of aspects such as specialization. When all the inefficiencies of small scale have disappeared, the firm may experience a period of constant returns scale, eventually, decreasing returns to scale may set in as the scale of production becomes very large.

A mathematical approach to production functions

A production function refers to the relationship between the output of a good and the input factors of production required to make the good.

In terms of formal notation a simple production function has the following general form:

Q =f(L,K)

Where Q is output

L is labour

K is Capital

This production function is typically used in the context theory of the firm. However, it is also possible to consider a nation's output as being dependent upon the various resources or inputs used to produce the output. In such a case the production function is termed as an aggregate production function and it would take the same general for as indicated above except that Q would now represent Gross National or Domestic Product and K and

L would refer to the whole of capital stock of the nation and the entire labour force respectively.

Production functions may take many algebraic forms. Economists usually work with homogenous production functions. A production function is homogeneous of degree **n** when inputs are multiplied by some constant, say **k,** and the output that results is a multiple of K^n times that original output.

Given a production function

$Q = f(L,K)$

If and only if

$Q = f(k,l,k,k) = K^n (f(l,k)$

Is the function homogenous. The exponent **n** denotes the degree of homogeneity. If n= 1 the production function is said to be homogeneous of degree 1 or linearly homogeneous. This, however, does not necessarily imply that the function is linear; it may or may not be.

A linearly homogeneous production function exhibits constant returns to scale. This is evident since the expression $K^n.f(l,k)$ when n – 1 is reduced to k.f(L,K) which imples that when inputs are multiplied by a constant k, output increase by the same proportion. One of the most popular forms of production function used in economic analysis is the Cobb-Douglas production function which is represented as follows:

$Q =A.L^a. K^b$

Where Q is output

A,a and b are constants

L is labour

K is capital

Recall that $AP_L = \frac{Q}{L}$

Where;

$AP_L=\frac{Q}{K}$

Where

AP_k is the average product of capital

Q is the level of output

K is the units of capital employed.

Given the Cobb-Douglas production function:

$Q = A.L^a . K^b$

$AP_k = \frac{Q}{L} = A.L^{a -1}.K^b$

$APK = Q = A. L^a .K^{b-1}$

The Cobb-Douglas production function is homogenous of degree a + b since the multiplication of L and K by the constant, K, will raise output by the proportion $K^{a+b.}$ If the exponents add up to unit the Cobb –Douglas function is linearly homogenous. This implies that is exhibits constant returns to scale. If the sum of the exponents is greater

than unity, the function exhibits increasing returns to scale. If the sum of the exponents is less than unity there are decreasing returns to scale.
The marginal product of labour is obtained by partially differentiating Q with respet L

$$MP_L = \frac{\delta Q}{\delta L}$$

In the Cobb- Douglas function

$$MP_L = aA.L^{a-1} k^b$$

Similary the marginal product of capital is given by:

$$MP_L = \frac{\delta Q}{\delta K} = bA.\ L^a\ k^{b-1}$$

COSTS OF PRODUCTION

Total, average and marginal cost

Fixed costs are costs that do not vary as output varies. They are costs associated with fixed factors of production and include items such as rent, rates, insurance, interest on loans, and depreciation. Fixed costs will be the same whether output is one unit or if output is 1,000 units. Fixed costs are also referred to as overhead costs or unavoidable costs. Depreciation, especially in capital –intensive industries usually constitutes a major item in fixed costs since the life of capital tends to be measured in economic rather than technical terms and machinery, for example. Depreciates even when not in use.
Variable costs are costs, that are related directly to output and include the wages of labour, the costs of raw materials, fuel and power. Variable costs are alternatively known as direct or prime costs.
Total costs represent the sum of fixed costs. (FC) and variable costs (FC)

TC = FC + VC

When output is equal to zero, total cots will equal fixed costs since variable costs will be zero. When production begins to increase total cots will continue to rise as variable costs increase since variable costs must increase at output expands. The breakdown of the total costs is shown below.

Output (Q)	**Total fixed costs (kshs)**	**Total variable costs (ksh)**	**Total costs (Ksh)**
0	50	0	50
1	50	20	70
2	50	30	80
3	50	35	85
4	50	45	95
5	50	65	115
6	50	110	160

Diagrammatically, the relationship between fixed costs, variable costs and total cots can be shown in figure 7.1. The shape of the total cost curve is explained largely by effect of increasing and diminishing returns to a variable factor. As output is increased, total variable cost at first rises at a decreasing rate as the firm experiences increasing rate as the firm experiences increasing returns to the variable factor. Subsequently, as the law of diminishing returns operates, total variable costs begin to rise at an increasing rate. The changes in total variable costs brought about by increasing and diminishing returns also imply changes in average variable costs.

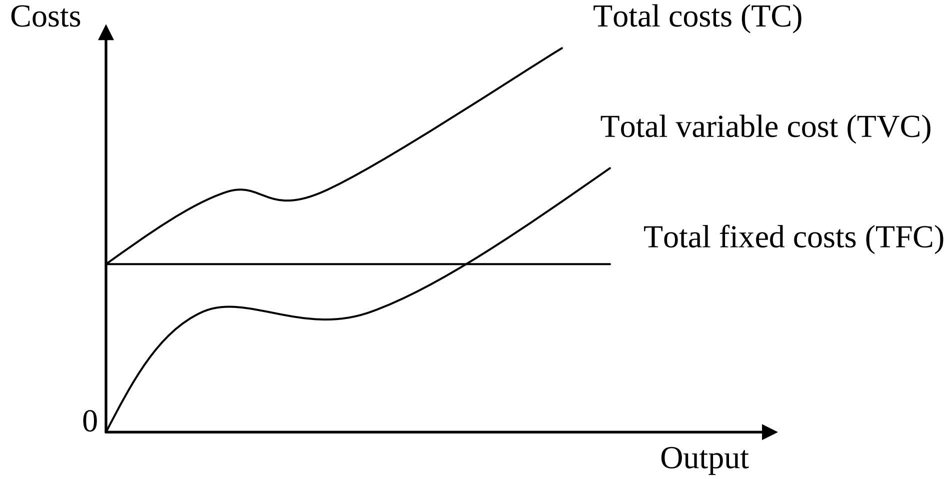


Figure 7.1: The relationship between fixed costs, variable costs and total costs

Average cost (AC) or cost per unit is the total cots of producing any given output divided by the number of units produced.

$$AC = \frac{TC}{Q}$$

Average total cost may be divided into average fixed costs and average variable costs in the same way as total costs were divided. At small levels of output, average cost will be high since fixed costs will be spread over a small number of units of output. However, with increases in output, average costs will tend to fall since each unit is carrying a smaller element of fixed cost and because for a time increasing returns to variable factors will be experienced. At a certain stage, however, diminishing returns will be encountered and average cost will begin to rise.

The average fixed cost (AFC) equals total fixed costs divided by output.

$$AFC = \frac{TFC}{Q}$$

Average variable cost (AVC) equals total variable costs divided by output.

$$AVC = \frac{TVC}{Q}$$

Average costs (AC) equals total costs divided by output. Average cost also equals average fixed cost (AFC) plus average variable cost (AVC).

$$AC= AFC + AVC$$

Marginal costs (MC) equals the change in TC or the change in TVC per unit change in output.

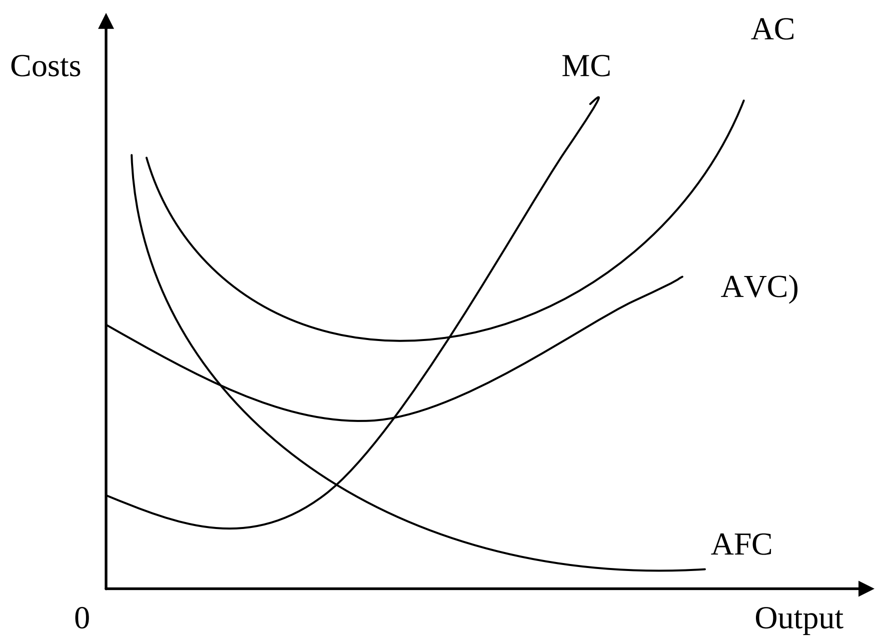
$$MC = \frac{\Delta TC}{\Delta Q}$$

The corresponding average and marginal cost for data in table 7.0 is obtained in table 7.2.

Table 7.2: the average fixed cost, average variable cost, average cost and marginal costs.

Output (Q)	Average fixed costs (Kshs)	Average variable cost (Ksh)	Average cost (Ksh)	Marginal cost (Ksh.)
1.	50	20	70	-
2.	25	15	40	10
3.	16.5	11.7	28.3	5
4.	12.5	11.3	23.8	10
5.	10	13	23	20
6.	8.3	18.3	26.7	45

Figure 7.2 shows the relationship between average and marginal costs.



In figure 7.2 it can be seen that the marginal cost curve cut the AC and AVC curves at their lowest points.

Mathematically, it can be shown that;

a. if the slope of AC< 0, then MC< AC
b. If the slope of AC = 0, then MC= AC.
c. If the slope of AC > 0, then MC > AC

Since the AC curve is u- shaped the slope of AC becomes zero at its minimum point. Hence, MC = AC at the minimum point of the AC curve.

This principle applies to any relationship between marginal and average figures. For example, assume that a particular student has an average of 60% after two tests. In his next test (marginal) he scores 50%. His average falls because the marginal score is below the existing average. However, if the next test he scores 80%, then marginal score now being greater than the average, would pull up the average.

A mathematical approach to cost functions

Total costs are usually expressed as a function of output (Q) only. A total cost production function can therefore be expressed in the following general form:

$$TC = f(Q)$$

Total costs are frequently represented by cubic functions. Cubic functions are multi-term functions where the highest power is three.

A general cubic cost function may take the following form:

$TC = a + bQ + cQ^2 + dQ^3$

(where a,b and d are >0, and c is < 0

The cubic function above represents the sum of fixed and variable costs where:

Fixed costs = a

Variable costs = $bQ + cQ^2 + dQ^3$

Average costs are obtained by dividing the total cost, the fixed costs and the variable costs by the level of output Q.

$AC = TC/Q = \frac{a}{Q} + b + cQ + dQ^2$

$AFC = FC/FC/Q + \frac{a}{Q}$

$AVC= VC/Q = b + cQ + dQ^2$

Average cost is the sum of average fixed cost (AFC) and the average variable cost (AVC) which can be demonstrated as follows:

$AC= \frac{a}{Q} + (b + cQ + dQ^2)$

Marginal cost refers to the instantaneous rate of change of the total cost function with respect to output Q

$MC = \frac{dTC}{dQ}$

For example, given the total cost function

$TC = 25 + 4Q -2Q^2 + 3Q^3$

$MC = dTC = 4-4Q +9Q^2$

$AC = 25 + 4-2Q + 3Q^2$

Example 1

The data below shows tabulation on the production of a hypothetical product.

Output(Q)Units	0	1	2	3	4	5	6	7	8
Total Cost(TC) Ksh	25	32	38	42	48	58	67	78	98

Using the above data, determine:

- **i)** Total fixed cost
- **ii)** Average variable cost when output equals 6 units
- **iii)** Marginal cost of the 3rd unit of output

Solution

i) **Fixed costs** are costs that do not vary as output varies. Fixed costs will be the same whether output is zero units, one unit or if output is 1,000 units. Therefore the fixed cost is 25 because even when the farm does not produce anything it incurs this cost.

Output (Q)	**Total cost**	**Fixed cost**	**Variable cost**	**Average variable cost**	**Marginal cost**
0	25	25	0	0	-
1	32	25	7	7	7
2	38	25	13	6.5	6
3	42	25	17	5.67	4
4	48	25	23	5.75	6
5	58	25	33	6.6	10
6	67	25	42	7	9
7	78	25	53	7.57	11
8	98	25	73	9.125	20

ii) From the table above average variable cost when output equals 6 units is 7

iii) Marginal cost of the 3rd unit of output is 4

Example 2

The total cost equation in the production of bacon some hypothetical factory is given as follows:

$C=1000+100Q-15Q^2+Q^3$

Where C = cost measured in shillings, while Q is the quantity measured in Kilograms.

i) Compute the total and average cost at the output level of 10 and 11 Kilograms

ii) What is the marginal cost of the 12th Kilogram?

Solution

Remember that a general cubic cost function takes the following form:

$TC = a + bQ + cQ^2 + dQ^3$

(Where a,b and d are >0, and c is < 0

The cubic function above represents the sum of fixed and variable costs where:
Fixed costs = a
Variable costs = $bQ + cQ^2 + dQ^3$
Therefore in our case the fixed cost is 1000 and the variable cost is $100Q-15Q^2+Q^3$

Output (Q)	Fixed costs	Variable costs	Total cost	Average cost	Marginal cost
0	1000	0	1000	-	-
1	1000	86	1086	1086.00	86
2	1000	148	1148	574.00	62
3	1000	192	1192	397.33	44
4	1000	224	1224	306.00	32
5	1000	250	1250	250.00	26
6	1000	276	1276	212.67	26
7	1000	308	1308	186.86	32
8	1000	352	1352	169.00	44
9	1000	414	1414	157.11	62
10	1000	500	1500	150.00	86
11	1000	616	1616	146.91	116
12	1000	768	1768	147.33	152

From the table above the total and average cost at the output level of 10 $is\ 1500\ and\ 150$ respectively

The marginal cost of the 12th kilogram is 152

Drawing that table is very tedious and time consuming; the simple way to handle these questions is as follows:

For the total and average cost at the output level of 10 and 11 kilograms you substitute Q with 10 and 11. Thus; for the output level of 10, total cost will be;
$C=1000+100Q-15Q^2+Q^3$
Total cost= $1000+100(10)-15(10^2)+10^3$
$=1000+1000-1500+1000$
$=1500$
Average cost=1500/10=150

For the output level of 11, total output will be;
$C=1000+100Q-15Q^2+Q^3$
Total cost=$1000+100(11)-15(11^2)+11^3$
$=1000+1100-1815+1331$
$=1616$

Average cost=1616/11=146.91

The marginal cost of the 12th kilogram is;

Differentiating the cost function we get;

$C=1000+100Q-15Q^2+Q^3$

$MC=dC/dQ=100-30Q+3Q^2$

$MC=100-30(12)+3(12^2)$

$MC=100-360+432$

$MC=172$

Economies of scale

In the long run, all the inputs into the production process are variable so the problems associated with diminishing returns to the variable factor do not arise. The law of diminishing returns, therefore, only applied to short run output decisions are concerned with diminishing returns given fixed factors of production, long run output decisions are concerned with economies of scale which are based on the assumption that all factors inputs are variable

Economies of scale are aspects of increasing size which lead to falling long run average cost. Economies of scale assist in the explanation of the trend towards larger production units in some industries. Economies of scale can be classified into internal and external economies of scale. Internal economies of scale are those factors which bring about a reduction in average costs as the scale of production of the individual firm rises, independently of what is happening to other firms.

External Economies of scale are those advantages in the form of lower average costs that a firm gains from the growth of the industry, that is, resulting from the simultaneous growth or interaction of a number of firms in the same or related industries. External economies are available to all firms in the industry no matter what their size is, economies can be illustrated in figure 7.3.

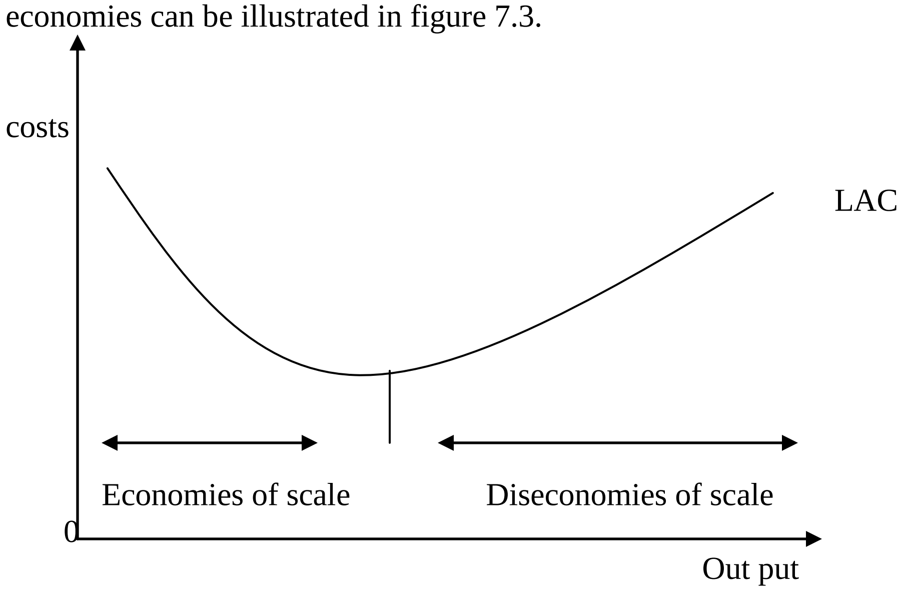


Figure 7.3: The concept of economies and diseconomies of scale

Internal economies of scale

a) Technical economies

These can be classified into the following categories:

i) **Increased specialization.** A larger scale of production implies that there is greater scope for the specialization of both labour and machinery. The repeated performance of the same actions means that labour can become very skilled. The breaking of the production process into many stages signifies that machines can be designed specifically for each stage. An example is in the motor assembly plant where many of the different stages in an assembly are completed using computer- controlled machines.

ii) **The economies of increased dimensions**. This is based on the principle that if the external dimensions of a container are increased, the cubic capacity increased more than proportionately, implying that the unit storage or transport cost of liquids or gases may fall as larger containers are used. This has led to the use of bigger tanks and chemical works. It has been reported by C.F. pattern that chemical engineers have developed a rule that if the capacity of a chemical plant is doubled, the capital cost is increased by only 52 per cent and the average fixed cost falls by 24% per cent.

iii) **Factor indivisibility**. Certain capital equipment has to be of a certain minimum size in order to justify manufacture. An automated car-assembly line is not a viable proposition for low volume output given that much of the capital equipment would be idle for long periods of time and its usefulness would hence not justify its cost. Through the use of indivisible capital equipment, larger firms can achieve lower average costs than small firms. The same principle can be applied to a research department and highly qualified specialists such as management consultants.

iv) **The principle of multiples**. Most industries make use of a variety of machines in their production processes with each machine likely to have a different capacity. In cases where the production process involves the use of different types of machinery, a large firm can arrange to have more of the machines with a small output and fewer of the machines with a high output thus achieving a high rate of utilization. Small firms, on the other hand, with only one of each type of machine, would find high capacity machines standing idle for most of the time.

b) Financial economies

Large firms have financial advantages in the sense that they are able to obtain finance at lower rates of interest than small firms. This is because they can often provide more collateral as security for loans than smaller firms. Additionally, the administrative costs of arranging a large loan are not twice those of arranging a loan half as large. Moreover, large firms have access to more sources of finance than the making of a new share issue on the stock exchanges.

c) **Marketing economies**

Marketing economies arise from the large scale purchase of inputs and from marketing and distributing the finished product in large quantities. Bulk buying facilitates the purchase of material by an enterprise on preferential terms including the obtaining of discounts. Large firms can also employ specialist buyers who have the knowledge and skill to enable them to economise greatly in the purchase of necessary materials. Additionally, selling costs per unit will generally be lower for large firms because these are spread over many units of output. Lower packaging costs per unit represent another marketing economy.

d) **Risk-bearing economies**

Large firms are better placed to withstand adverse trading conditions since they can benefit from the law of averages. Thus, large firms can operate with small stocks in proportion to sales since the variation in orders form individual customers will tend to offset each other. Risks of trading can be reduced by large firms through the diversification of products and markets.

Internal diseconomies of scale

Increasing the size of an individual firm beyond a certain scale can lead to rising average costs. This is fundamentally because of management difficulties and rising prices of inputs.

Management problems arise because:

i. As the size of departments in an organization increase, the task of coordination becomes more difficult.

ii. Despite the existence of a hierarchy of authority in large firms, the task of control, that is, of ensuring implementation, is extremely difficult in practice.

iii. In firms of large size communication may be problematic in that it is difficult to ensuring implementation, is extremely difficult in practice.

iv. In firms of large size communication may be problematic in that it is difficult to ensure an effective vertical and lateral line of communication. Communication networks are generally more complex in large organizations with the associated greater likelihood of communication breakdowns.

v. The maintenance of morale is more difficult in large organizations because individual workers in large organizations may feel unimportant to the firm and often do not identify with the firm's objectives.

An additional source of internal diseconomies of scale is increases in the prices of inputs since as the scale of production increases, the firm will increase the demand for inputs like labour and transport and this may lead to the bidding up of the prices of certain inputs.

External economies of scale

External economies of scale or economies of concentration a rise form the simultaneous growth or interaction of a number of firms in the same or related industries. External economies of scale are particularly important in policies of industrial location and external economies are available to all firms in the industry irrespective of their size. External economies of scale arise form the following factors:

i. The creation of a labour force skilled in various techniques used in the industry. Special training courses may be offered by local colleges geared to the particular needs of industries in a particular area.

ii. The development of subsidiary industries catering for special needs of a major industry or the provision of ancillary services. Examples would include firms specializing in the wholesaling, marketing or processing of waste products from local industries.

iii. In the case of disintegration which refers to the tendency for individual firms to specialize in a single process or in the enable individual firms to obtain their component in an industry that would enable individual firms to obtain their components and other requirements at relatively low cost since they are being mass produced for the industry.

iv. Co-operation between firms can take the form of joint research centres and the formation of trade societies which is easier in the case of localized industry.

v. External economies may arise through the provision of commercial facilities by service industries in the area that develop a special knowledge of the needs of the industry. Examples include the development of specialized banking, insurance and transport services tailored to the requirements of a particular industry.

External diseconomies of scale

External diseconomies of scale may arise because a shortage of various inputs used in the industry may develop, leading to an increase in the cost of those inputs. Thus, for example, an increased demand for raw materials may bid up the prices of raw materials and cause their prices to rise. Heavy localization of industry may make land for expansion scarce and therefore more expensive to rent and purchase increased congestion could also lead to higher transport costs.

The relationship between long rung and short run cost

It should be recalled that in the long run there are no fixed factors of production and expansion can occur through the acquisition of new plant and equipment while the scale of operations can be reduced through the replacement of plant and equipment as they wear out or disposal can occur through the sale of assets. The relationship between long run and short run cost curves can be explained in figure 7.4

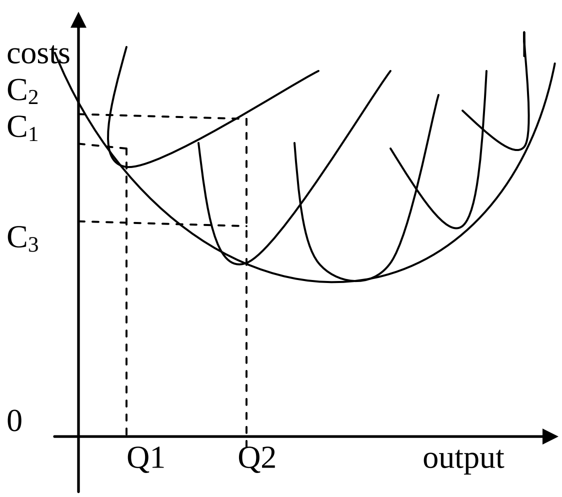


Figure 7.4: The relationship between short run and long run average cost curves.

In figure 7.4, the original short run cost curve can be said to be represented by SAC_1. If a firm succeeds in increasing its scale of production the short run cost curve becomes effective for each particular scale of production and these short run curves can be represented by SAC_1, SAC_2, SAC_3, SAC_4 and SAC_5. Each of these short run cost curves represents a different stock of capital and other fixed factors of production. If one assumes that the firm will always choose the size of plant that minimizes the cost of producing any level of output, it will be economically rational for the firm to increase its scale of production as the demand for its product increases. Assuming that the firm is initially operating on the short run cost curve $SAC_{1\ \text{and}}$ producing an output of Q_1 at an average cost of $C_{1.}$ Further assume that there is an increase in demand that leads to the firm increasing its output to Q_2. If the existing size of plant is maintained, this would raise average costs to C_2 . However, by increasing the size of plant and moving to the cost curve SAC_2 , the output Q_2 could be produced at a lower average cots given by C_2. The movements to the cost curves $SAC_{3,}$ SAC_4 and SAC_5 can be explained by using the same reasoning in that instead of moving up an existing cost curve, lower costs of production can be achieved by moving to the cost curve of a plant with greater capacity.

The long run average cost curve consists of a series of points on different short run cost curves and the point on the long run average cost curve represent the lowest costs attainable for the production of any given level of output. The long run average cost curve can therefore be described as an "envelope curve" to a series of short run average cost curves.

The optimum size of a firm

The optimum size of a firm is the best or most efficient size of a firm when the long run average cost of the firm is at minimum. The optimum size can be shown in the previous diagram by the output Q*. At this size there will be no motive for further expansion since at any other size large or smaller the firm would be less efficient. Between one industry and another the optimum size will vary since small firms may be more efficient for some purposes and large ones more efficient for other purposes. For an optimum firm to exist factors of production must be combined in the optimum proportion and the firm must have reached a size where it can full economies of scale.

There are several difficulties with the concept of the optimum firm. Firstly, it is difficult to decide in actual cases whether a firm has reached the optimum size or not. Secondly, it has been suggested that there may be more than one optimum, a low and a high optimum with the expansion from the low optimum to the high optimum being difficult because any intermediate point between the tow optima would be considered less efficient. Thirdly, the optimum in any industry may itself vary form one period to another as a result of changing conditions and the development of new techniques. Thus, for example, a change in relative prices and the efficiency of factors will affect the optimum proportion in which the factors are combined. Fourthly, a major difficulty confronting any firm striving to reach its optimum size is that the main divisions of the firm may have different optima. Thus, for example, the optimum administrative division may not suit the optimum technical division which may require an administrative department of either greater or lesser than the optimum of the other departments.

Types of mergers.

Vertical integration

Vertical integration is said to occur when a merger takes place between firms engaged indifferent stages of the production process. Thus, for example, a tyre manufacturer can acquire rubber plantations. Backward integration is said to take place when the movement is towards the source of supplies. Forward integration takes place when the movement is towards the market outlet as, for example, in the case of large oil companies taking control of petrol stations.

Horizontal integration

Horizontal integration is said to take place when firms engaged in the production o the same kind of good or service are brought under unified control. An example would be the amalgamation of several motor manufacturers.

Diversification

Diversification is said to occur when firms that produce goods or services that are not directly related to each other combine. An example would be the merger of a firm producing fertilizers with a manufacturer of paint. The main aim of diversification or conglomerates is to reduce the risk of trading.

The survival of small firms

Small firms continue to survive for the following reasons:

i. A demand for variety that cannot be met by mass production especially in industries like clothing and footwear.
ii. Many owners of small firms have no ambition to grow large because they do not want to sacrifice their independence and control.
iii. Personal contact with customers is important in many industries like accountancy and architecture
iv. The size of the firm may be limited by extent of the market since a firm can only grow in size if this is permitted by the market. The market for luxury items for example is limited by income and wealth.
v. Firms may want to avoid the rising costs that arise from diseconomies of scale.
vi. There is a tendency for mass production industries to disintegrate into a large number of specialist firms.

MARKET STRUCTURES

An introduction

Business organisations operate in two types of markets. One for **factors of production** such as labour, and one for **output.** This chapter deals with the output markets. The structures of the markets in which firms operate may vary. The implications of these variations are vital for an understanding of the environment in which a business operates. In this chapter we consider the pricing and output decisions by firms, which operate in different market structures. These different forms of market structures are characterised as perfect competition and oligopoly. The latter three forms of market organisation fall within the realm of imperfect competition. These market structures are identified by economists in order to systemize and organise their analysis. In the real world, such a sharp distinction between different market structures may not exist since firms may exhibit elements of more than one market form making it difficult to strictly classify them into any of the above categories.

A **market** in economics refers to a situation or context where potential sellers of a commodity are brought into contact with potential buyers and a means of exchange is available. There does not need to be a physical entity corresponding to a market. Thus, for example, a market may be created over a telephone or through the Internet.

Perfect competition

A market is said to be perfectly competitive if the following conditions hold:

i) Where there are many buyers and sellers so that the actions of an individual cannot have a significant impact on the market price. This is because each buyer or seller is too small in relation to the market to be able to affect the price of the commodity by his or her own actions.

ii) Where the product is homogeneous such that buyers do not have a preference for the product of any particular producer. The buyer cannot distinguish between the output of one firm and that of another, and as such is indifferent as to the particular firm from which he or she buys.

iii) Where there's perfect mobility of factor s of production such that producers can respond to price signals and there is free entry and exit to and from an industry. In the case of perfect mobility of factors of production no input required in the production of the commodity is monopolised by its owners or producers. Firms can enter or leave the industry without much difficulty in the long run. There are no patents or copyrights and considerable amounts of capital are not necessary to enter the industry. Established firms do not have a lasting cost advantage over new entrants owing to the size or experience.

iv) Where consumers, factor owners and producers have a perfect knowledge of the prevailing market conditions such that no price differentials will exists. Perfect knowledge of the prevailing market conditions such that no price differentials will exists. Perfect knowledge of the market is in relation to present and future prices, costs and economic opportunities in general. Perfect knowledge of the market implies that consumers will not pay a higher price than necessary for the commodity and price differences will be quickly eliminated. Given the perfect knowledge of present and future prices and costs. Producers will know exactly what level of output to produce.

v) Economics agents should be rational such that consumers seek to maximize their utility and producers and resource owners seek to maximize profits.

In reality to known market satisfies the conditions of perfect competition. Examples of markets that closely **appropriate** the conditions of perfect competition include world agricultural markets for commodities such as coffee and foreign exchange markets.
The model of perfect competition nevertheless provides a useful theoretical benchmark against which one can assess the degree of competition in real world markets. It also enables us to contrast the behaviour of firms in markets where competition shows some of the characteristics of being perfect.

In perfect competition since each unit of output is sold at the same price, both the marginal and the average revenue are constant. This can be illustrated in Table below.

Price (A) (Ksh)	**Quantity Demanded**	**Total Revenue (Ksh)**	**Marginal Revenue (Ksh)**
20	1	20	20
20	2	40	20
20	3	60	20
20	4	80	20
20	5	100	20
20	6	120	20
20	7	140	20
20	8	160	20

The above table illustrates that as the quantity demanded increases, the price remains unchanged. This implies that each additional unit sold increases the total revenue by an amount equal to its price.
The demand curve for a perfectly competitive firm is infinitely elastic. The price is determined by the market and cannot be influenced by an individual firm whose output is too small relative to total industry output. The market price applies to each firm and is determined by the market forces of demand and supply. The **industry** demand curve, unlike the individual demand curve, slopes downwards because if the industry as a whole reduces its price it will sell more. Should the price rise or fall then the new price will appl to every firm in the industry. This implies that in perfect competition a firm is a **price taker** and can sell any amount of the commodity at the established market price. The relationship between the individual firm's price and the market industry price an be shown in Figure 8.2.

In Figure 8.2 the demand curve for the product of an individual firm is perfectly elastic whereas the market demand curve for the output of the industry will be a normal shape in that it slopes downwards from left to right. DD and SS are the industry demand and supply curve respectively.

The conditions for profit maximisation

Profit is defined as the difference between total revenue received by the firm and the total costs it incurs. Profit maximisation requires that the following tow conditions are satisfied.

i) The necessary condition

According to this condition profits are maximised at the level of output where:

Marginal Revenue = Marginal Cost

To intuitively understand why this is so, consider a situation where marginal cost is greater than marginal revenue. In such a case, the costs of producing an additional unit of output are greater than the revenue obtained and hence profits are lower than marginal cost opportunities exist for increasing profits by expanding output.

It can be proved mathematically that marginal revenue must equal marginal cost at the profit maximizing level of output

Given II = TR-TC

Where II = Profit

TR = Total Revenue

TC = Total Cost

To miximise profits. $\frac{DII}{dQ}$ must equal zero (see appendix II for maxima and minimal).

$$\frac{dII}{dQ} = \frac{dTR}{dQ} - \frac{dTC}{dQ} = 0$$

$$\frac{dTR}{dQ} = \frac{dTC}{dQ}$$

But $\frac{dTR}{dQ}$ = MR and $\frac{dTC}{dQ}$ = MC

Therefore, MR = MC at the profit maximising level of output.

ii) The sufficient condition states that the slope of the marginal revenue curve must be less than the slope of the marginal cure at the point where they are equal. This implies that the marginal cost curve must cut the marginal revenue curve from below. This condition ensures that marginal cost is less than marginal revenue at points below the profit maximising level of output. These conditions may be illustrated in Figure below.

The first condition for profit maximisation (necessary condition) is that the firm should produce at that level of output where marginal cost (MC) = marginal revenue (MR).

In Figure below this is the case at the two output levels Q_1 and Q_2. Output level Q_2.is the most profitable. At output levels below Q_2.profits can be increased since an increase in output adds more to total revenue than it does to total costs. At output levels greater than Q_2.the reverse applies and profits can be increased by lowering output. At output Q_1 although MC should be rising and cut MC from below.

The necessary and sufficient conditions determine the rate of output at which profits are mximised but they leave open the question of whether profits are positive or negative.

The equality of marginal cost and marginal revenue may well be at the level of output where losses are minimised. Profits will only be positive if total revenue exceeds total costs as in the case of Figure below Even though the above analysis has been carried out assuming perfect competition, the same conditions would apply if a firm is operating in imperfect competition.

Monopoly

A **monopoly** is a market structure where production is under the control of a single supplier. Since the monopolist is the sole source of supply in a market, the demand curve is also the industry demand curve. This implies that the monopolist faces a downward sloping demand whereby if the monopolist wants to sell more he must reduce the price.

In the monopoly market structure, average revenue is not the same as marginal revenue. Marginal revenue is less than average revenue because wen the monopolist wants to sell more he must reduce the price, not only on the extra units sold, but also on all the earlier units.
This relationship can be illustrated in Table 8.2

Table 8.2 The revenue position of a firm operating under imperfect competition or monopoly.

Price (AR) (Ksh)	**Quantity Demanded (units)**	**Total Revenue (Ksh)**	**Marginal Revenue (Kshs)**	**Price elasticity Of demand**
20	1	20	20	$E_D > 1$
18	2	36	16	$E_D > 1$
16	3	48	12	$E_D > 1$
14	4	56	8	$E_D > 1$
12	5	60	4	$E_D > 1$
10	6	60	0	$E_D = 0$
8	7	56	-4	$E_D < 1$
6	8	48	-8	$E_D < 1$
4	9	36	-12	$E_D < 1$
2	10	20	-16	$E_D < 1$

In Table 8.2 average and marginal revenue start off as equal but thereafter as the firm sells at a higher level of output, marginal revenue falls further below the average revenue. Both the average and marginal revenue decrease as more is sold. Both the average and marginal revenue decrease as more is sold. Figure 8.9 illustrates the relationship between the average revenue and marginal revenue in a monopoly.

Mathematically, Marginal Revenue has twice the slope of Average Revenue.
This can be shown as follows:

If $\quad P \quad = \quad a - bQ$

And $\quad TR \quad = \quad P \times Q$

$$\text{Then} \quad TR = aQ - bQ^2$$

$$\text{Since} \quad MR = \frac{dTR}{dQ} = a-2bQ$$

Therefore, MR has twice the slope of AR.

Sources of monopoly power

The following factors may constitute barriers to envy and hence could be a source of monopoly power.

- **i.** **Legal barriers**. These may take the form of statutory monopolies or patents. Statutory monopolies are those established b an Act of Parliament. For example. Nationalized industries. A companies' products may also be protected from competition from new firms for a certain period of time**.**
- ii. **Product differentiation barriers**. These may, for example, take the form of advertising and branding whereby an existing monopolist may exploit his position as a supplier of an established product which the consumer may be persuaded to believe is better. The initial advertising costs of establishing a new brand may be considerable.
- **iii.** **Economies of scale barriers.** These would arise where existing firms are already operating on a vast scale of production and enjoying technical economies of scale. New entrants into the market would have to achieve a substantial market share before they could gain full advantage of potential economies of scale.
- iv. **Transport cost and tariff barriers.** Some firms may enjoy local monopoly positions arising from the ability to sell more cheaply in heir own localities than other firms. Such firms can therefore raise their prices in their local markets above their production costs by an amount that does not exceed the transport costs of firms in other localities with similar production costs. Tariffs may protect the domestic market from competition from foreign producers by artificially raising the price of the foreign product in the domestic market.

Price discrimination

Price discrimination is the practice of changing different prices for the **same** commodity. Price discrimination may sometimes be referred to as discriminating monopoly.
Goods and services may be differentiated by time or place so that buyers cannot easily shift to the lower priced commodities. Price discrimination could be based on **individuals, locality** or **use**. When different prices are charged from one person to another, the price discrimination is referred to as **personal** price discrimination. When the price vary from one locality is know as **local** price discrimination. Price discrimination may also be based on **use.** Such that different prices are charged according to the use to which the commodity is put. For example, in may countries electricity for industrial use is sold at lower prices than electricity for domestic use.

Conditions that favor the effective use of price discrimination

i. The monopolist should be able to keep the various sub-markets separate so that it is not easy to transfer units of the commodity from one sub-market to the other. Markets may, for example be separated by distance or tariff barriers.
ii. Price discrimination can only be practiced where there is imperfect competition. This is because under conditions of perfect competition, firms are price takers and cannot charged different prices for the same commodity. This is partly because consumers have a perfect knowledge of the market.
iii. Discrimination can be practiced on the basis that different groups of consumers have different price elasticity of demand. The discriminating firm can therefore charge a higher price to higher income consumers who have a low price elasticity of demand and are less responsive to changes in prices. On the other hand, a lower price can be charged to lower income consumers who have a higher price elasticity of demand and are thus more sensitive to price changes. Price discrimination can be applied on this basis especially when consumers have certain strong preferences and prejudices. For example, higher income groups may have a strong, though irrational feeling that they are paying a higher price for better quality commodities. Such individuals may also prefer to purchase their commodities in fashionable, but relatively expensive areas.
iv. The nature of the commodity is also important for price discrimination to be effective. Direct services such as restaurant services, medical services, hair cuts and so on are particularly suitable for price discrimination.
v. Government regulations may sometimes encourage price discrimination. For electricity and water for industrial and domestic users.
vi. The purchase of special orders may also provide sellers with an opportunity to charge different prices. This is because in this situation customers cannot readily compare the prices being charged to different customers.

Monopolistic competition

Monopolistic competition is a form of imperfect competition which lies between the extremes of perfect competition and monopoly and includes elements of both. Monopolistic competition is common in the service sector, for example, in the case of restaurants and hair dressers.

Monopolistic competition has the following features from the **monopoly:**

i. A firm in this market structure has a certain monopoly power over its own brand because no one else can produce it, although the products are differentiated substitutes.
ii. As in monopoly, firms in a monopolistic competition face a downward sloping demand curve.

Monopolistic competition has the following features from **perfect competition:**

i. There are many buyers and sellers in the market.
ii. There is freedom of entry and exit to and from the industry.

In monopolistic competition, each producer sells a product which is slightly different from his competitor's products and will attempt to emphasis these differences. In cases where there are no real differences, the producer will attempt to create them through appropriate packaging and advertising. The process of creating differentiation is aimed at creating brand loyalty.

In monopolistic competition there are either no barriers or low barriers to entry. Firms entering the market do not require a large scale capital investment.

Although there are many firms in monopolistic competition, each firm is selling a **differentiated** product and hence has some limited control over the price. Firms in monopolistic competition are therefore price makers since they can raise heir prices without losing all their customers and they have to reduce their prices in order to sell more.

The firm's demand curve in monopolistic competition will be relatively elastic because the commodities sold by its competitors will e relatively close substitutes.

Oligopoly

Oligopoly refers to a market structure dominated by a few sellers. The entire output in an oligopolistic industry is produced by a few large firms and the contribution of each firm is sufficiently large to be significant in the market. Examples of oligopolistic firms are usually found in the oil, motor vehicle and cigarette industries.

In addition, when the number of competitors in the market are few, each seller becomes acutely aware of how rival firms are likely to react to any change a particular firm may make,especially relating to price.

Oligopoly is unique as a market structure in that the firm's pricing and outdoor decisions will also incorporate the **perceived** or **expected** reaction of competitors. This implies that the policies of firms in oligopolistic structures are **interdependent.**

This interdependence of oligopolistic firms helps to explain one of the major characteristics of oligopolistic firms markets-that of **price rigidity.** Prices tend to be sticky in the sense that they are unlikely to change very often.

NATIONAL INCOME

Definition of national income

National income is a measure of the total **monetary** value of the **flow** of **final** goods and services arising from the productive activities of a nation in any one year. Although money value is used, the concern of economists is with **real** things.

Various concepts of national income

i. **Gross Domestic Product (GDP).** Total monetary value of all final goods and services produced within the geographical boundaries of a country. The term'gross implies that no deduction for the value of the expenditure goods for replacement purposes (depreciation) is made. Since the income arising from the investments and possessions owned abroad is not included, only on the value of the flow of goods and services produced in the country is estimated. Hence the word'domestic' is used to distinguish it from Gross National Product. The output considered in this case is that produced by all individuals in country, irrespective of their citizenship.

ii. **Gross National Product(GNP):** Total monetary value of all goods and services produced by the nationals or citizens of a given country. Some of this output may be produced by nationals engaged in productive activities abroad.

GNP = GDP + Net factor income from abroad.

Net factor income from abroad refers to the difference between the income accruing to domestic residents arising from productive activities abroad less income earned within the domestic economy accruing to non-residents. This distinction arises because domestic production may be carried out with the assistance of inputs belonging to the residents of other countries while it is also possible for the residents of a given country to be engaged in productive activities abroad.

The difference between GDP and GNP will be greater in those countries which allow a high level of foreign investment developing countries where multinational corporations have invested heavily. These developing countries have, however, not invested considerably abroad themselves. Thus their estimated GDP is usually greater than their GNP.

iii. National disposable income

National Disposable Income: = national income +(-) net transfer receipts or payments. This concept becomes useful where a country receives substantial transfers from abroad either to the government or to individual residents, or alternatively where the residents of a given country send out substantial remittances. The concept of National Disposable Income, therefore, measure the aggregate resources available to a nation for saving or consumption.

iv. Nominal versus real national income

National income is concerned with changes in the volume of output. But, to obtain uniformity, national income is measured in money value. However, a change in money value can occur because of a change in either quantity or price. Thus money value can change while leaving the volume of goods and services constant. So an inflational deduction known as 'Stock Appropriation' has to be carried out.

Real national output, therefore, refers to the value of total output measured in 'constant prices' (measured in the prices of a base year). This implies that the general rate of inflation is deducted so as to record the real command over resources.

Nominal nation output describes the measurement of total output in the current prices. Price indices are used to convert nominal values, whose measurement is in the current year prices, to real values by keeping prices at a selected base level.

Year capita income
Per capita income is the national income divided by the total population of the country. It represents the average income of the people in a given year (Income per Head).

$$\text{Per capita income} \quad \frac{\text{National income}}{\text{Total population}}$$

Uses of national income statistics

National income statistics may be useful in the following ways:

To indicate the standard of living
In economics, the standard of living can be taken to refer to a measure of people's living conditions, measured by their material well being. The ability of national income to indicate the standards of living of the country depends on both presenting the statistics in the right way and on our understanding the severe limitations of this concept as a measure. The problems involved with using national income may suggest that it is used because it is the best index we have, not because it is accurate.

Limitations of using national income statistics to indicate the standard of living

a) Other forms of wealth: National income statistics consider the flow of wealth created by production, around the economy. But a substantial proportion of wealth does not flow, for example, property and houses, rare paintings, stocks and shares. Thus national income statistics do not take account of the well-being gained from for example owning a valuable painting or other no-economically measurable benefits that contribute to our welfare such as leisure, long holidays, job satisfaction, health , good friends, and pleasant surrounding's.

b) **Inflation**

A rise of GNP in money terms, may be accompanied by fall in real terms (in terms of how many goods and services we are receiving) because statistics are recorded in money terms, and the value of money itself can change. Inflation may therefore distort national income estimates. In measuring standards of living we are interested in how many goods and services we will receive and not in receiving more money that will buy less.

c) **Population**

The measurement of the total value of goods and services produced in a year is itself insufficient to make conclusions about standards of living. Countries may have similar levels of GDP but considerably different levels of GDP per head. In additions, a country may have a rising GDP per head, not because the real output is rinsing, but due to a population decline.

d) Distribution of income

The use of income per capita as a measure of living standards presupposes that all people are receiving the same proportion of the national cake. Despite taxation and welfare distribution, the highest income is normally concentrated in the hands of about 1% of he total population. So, an increase in national income may be distributed among a small proportion of the population.

e) Working hours

An increase in national income may be the result of longer working hours and inferior working conditions perhaps also accompanied by longer journeys to work.

To compare the standards of living in different countries

National income statistics are often used to compare the standards of living in different countries. GDP statistics also form the basis of major international contributions and receipts. GDP forms the basis on which the IMF quotas are fixed, and through that the members' voting power, the amount of foreign exchange that can be drawn from the IMF and the allocation of Special Drawing Rights (SDRs).

Limitations of using national income statistics to compare standards of living in different countries.

(a) Different currencies

Since figures are expressed in different currencies, they need to be converted into a common denominator. It would appear simple enough to convert all the figures into one currency valuation, but this is not as straightforward as it may seem. A major problem is that the major trading nations have floating exchange rates, which fluctuate dramatically. Moreover, the rate may not accurately reflect the internal purchasing power of a currency. This implies that the market rate of exchange is not necessarily the ideal measure of the relative values of the goods and services consumed in each country.

(b) Different definitions

The goods and services included in the definitions of national income differ from country to country. Thus for example, the former Eastern European countries excluded the value of non-material services such as public administration, and they also included defense and personal and private services. When comparisons are made, account must be taken of these different definitions to avoid rendering them meaningless. The subsistence sector constitutes a considerable proportion of the economy in developing countries. Although developed countries such as the United Kingdom exclude the subsistence sector, for developing countries to do this would significantly undervalue both their national product and their standards of living. The problem here is one of the product boundary. In addition, goods considered illegal in some countries may not be so in others.

(c) Distribution of income

Although income per capita may be similar, standards of living may differ considerably because of differences in income distribution. A country may have a high average income per head, which is earned by a minority while the majority of the population is poor. This is the case for example, in some of the oil-producing states of the Middle East.

(d) Different needs and tastes

In using per capita income as a measure of the standard of living, the standard of living of one country should not be compared to or assessed in terms of what another country finds valuable. For example, the fact that people in Northern countries spend more money on heating does not make them better off. This implies that before a comparison can be made, we must be certain we are measuring things that mean the same thing to different people.

(d) Income in relation to effort

An increase in national income may be the result of longer working hours, inferior working conditions or longer journeys to work. Thus an increase in income may be accompanied by an decline in the standard of living. Also a decline in income may occur because of increased leisure, although the decreased working hours may compensate for this decline in income to some extent.

To assess the rate at which a nation's income is growing

Growth is a major government objective and so it is essential to be able to measure it. **Economic growth** may be defined as an increase in a country's productive capacity, identifiable by a sustained increase in real national income over a period of time. **Economic development,** on the other hand, is associated with a rise in real national income and is also accompanied by major structural changes in the economy, such as a shift from agriculture to manufacturing. National incomestatistics for the purpose of assessing the rate at which a nation's income is growing have limitations such as changing definitions and inflation. It is therefore difficult to make comparisons from period to another and also intercountry comparison of economic growth.

To assist the government in planning the economy

Without a record of the total National Product and its components, the government cannot plan the running of the economy and check whether its economic objectives are being achieved. National income statistics play a role in the planning of both short and long run government policies. They show trends in consumption and investments, changes in employment, inflation and economic growth due to changes in national income. They enable the government to assess what policy measures to take to correct adverse trends such as low investment. They are useful in repairing the budget so as to determine expenditure in appropriate sectors and formulate sectors policies,. These statistics also facilitate a redistribution of income in the economy as government policies are formulated to remove inequalities.

To forecast future trends

As well as recording past performance, the national income statistics may be used to forecast future performance. Proposed changes in the budget, for instance, will be compared against the likely outcome. O the basis of these statistics the future strategy is planned. National income statistics can help to better understand the manner and pattern of economic activity, and to provide an empirical database upon which economic models are built. Forecasts can assist institutions financing different sectors.

Business community

The business community can use some of the data provided by national income statistics to understand market trends so as to invest appropriately, For example, national income sectoral data can assist in showing which sectors are growing and which are declining.

Methods of measuring national income.

We now consider the methods of measurement of national income, outlining the major adjustments that have to be made so that the statistics are accurately recorded and the estimates by different methods are equivalent. Each time a commodity is produced and sold we can say that:

A. The value of the commodity is equal to the purchaser's **expenditure** on it.

B. The same sum of money will be received as **income** by the different individuals who contributed to the production at some stage.

C. The value of the commodity sold will have resulted from the **value added** to it by successive stages of production.

The above implies the total monetary value of goods and services produced and sold during a particular year can in theory be calculated by:

A. Adding together market expenditures by final consumers.

B. Adding the income received by different factors of production which have contributed to the production.

C. Determining the value of each firm's contribution to output or its "value added".

These alternative measures provide a basis for the expenditure, income and output methods which are really **three** ways of arriving at the **same** total.

Measurement by the income method

The **income approach** takes national income as the sum of all incomes earned by factors of production in the economy: included are **personal incomes** which have been earned for services rendered and in respect of which there has been some corresponding value of output, **income as benefits** such as rent-free housing should be imputed as income from employment both in private and public sectors, the **gross trading profits of companies** and **gross trading surpluses of public corporation** and the **government,** imputed rent(rent broadly taken to include income from land or property such as housing) and income from **subsistence production.**

Measurement by the product or value added approach

This constitutes the most direct approach. Using this approach, national income is found by adding up the value of all-final goods and services produced by firms during the year. National income is assumed to be the total of the vale added to output by all enterprises in economy. The total or value added by different stages of production equals the final value of output of a final good.

Measurement by the expenditure method

This approach arrives at National income by adding together expenditure on all final goods and services in the economy. Total expenditure is broken into 5 broad categories. Consumer expenditure government spending, private investment expenditure imports and exports. This can be represented by the following identity.

$$\text{Expenditure } C + G + I + X - M$$

Where C is the consumer expenditure, G is government expenditure I is private investment expenditure, X exports and M is imports administration, law and order and defence. Subsistence output is actually pay himself any money.

MONEY AND BANKING

In modern economies, money is used as a means of paying for goods and services, and paying for labour, capital and other resources. Money “oils the wheels” of economic activity by providing an easy method for exchanging goods and services. **Money** refers to anything which is widely acceptable in payment for commodities and in settling debts, not for itself but because it can e similarly passed on.

The functions of money

A. Money as a means of exchange

This is a vital function of money in an economy because without money, the only way of exchanging goods and services would be by means of barter, which implies a direct exchange e of one commodity for another. The economies we live in are monetary economics in which most of the goods and services produced are exchanged vial the intermediary of money, rather than through barter. Whenever money is used to pay for goods and services or for the purpose of settling transactions it is functioning as a medium of exchange.

B. Money as a unit of account

Money should be able to measure exactly what something is worth. Money should provide an agreed standard measure by which the value of different goods and services can be compared. This implies that the functions of money in the economy could be to establish a cocoon unit of measurement by which the relative exchange values or prices of goods and services can be established. Money provides a convenient means of measuring economic phenomena.

C. Money as a standard of deferred payment

The function of money in this respect allows people to delay paying for goods and services or settling a debt, even though goods and services are being provided immediately. Money acts as a standard of deferred payment whenever firms sell goods”on credit”. Money facilitates the extension of credit by specifying the unit of future payment.

D. Money as a store of value.

This function of money arises when instead of spending money a person decides to store his or her wealth in the form of money rather than other forms of wealth such as property or financial assets(for example, shares). This implies that the purchasing power of money is transferred into the future although this may be eroded by inflation. Money, therefore, permits income recipients to postpone consumption or save on the basis that the money can be used for future consumption.

Money can therefore, be defined as those items which fulfil each of the functions mentioned above. The form money takes therefore depends on the actual items used at any point as a medium of exchange, a unit of account, a standard of deferred payment, and a store of value.

The effects of inflation on the functions of money

Inflation refers t the persistent increase in the general price level. Inflation inhibits the ability of money to perform its functions effectively in the following ways.

1) As a means of exchange

When inflation is extremely high people may prefer not to use money at all. They may instead revert back to the barter economy or use money alternatives such as gold or a foreign currency whose value will increase against the currency of hyperinflation economy.

2) As a unit of account

Inflation will contribute to the variations of relative values of different commodities over time since individual goods or services do not rise in price by the same proportion. This reduces the stability of money as a unit of account.

3) As a standard of deferred payment

In periods of inflation, lenders lose while borrowers gain because the real value of the debt declines. Lenders may, therefore, only accept to lend money at very high rates of interest in order to compensate for the fall in the real value of the loan.

4) As a store of value

During the periods of inflation, although money retains its nominal value it loses some of its purchasing power. His implies that the real value of money declines. The loss of this value may cause people to store their wealth in a variety of assets such as property or paintings which they do not expect to lose value.

The role of central banks

Central banks have traditionally the following roles in the economy of a country:

1. Central generally formulate and implement the monetary policy in a given country. This functions is usually directed to achieving and maintaining price stability. In this regard Central banks have sometimes issued guidelines on credit and interest rates. In addition, they have often been agencies of governments for issuing and underwriting of government borrowing instruments such as treasury bills. They, therefore, play a vital role in controlling the level of money supply in a given country.

2. Central banks have a role of encouraging liquidity, solvency and proper functioning of he financial system. Central banks thus sometimes inspect commercial banks and other financial institutions and in this way become principal advisors on whether or

not to issue or renew licences of financial institutions. In addition, authorised dealers in the banking markets are usually supervised by the central bank.

3. Central banks usually formulate and implement a country's foreign exchange policy. In countries that have exchange control, the central bank may be responsible for enforcing such controls. A related aspect is the management of a country's foreign exchange reserves, which usually conducted by the Central Bank.

4. Central banks also usually act as bank advisers and fiscal agents of the government.

5. Central banks issue currency notes and coins in their respective countries. In this regard they often took over from currency boards which originally had the function of issuing currencies.

Money and capital markets

Financial markets

Refer to all trades that result in the creation of financial assets and financial liabilities. They are concerned with the relationship that subsists between the suppliers and demanders of funds, including the intermediate institutions, which are the demanders and suppliers of funds. The three types of financial markets are the **money market,** the **capital markets** and the **foreign exchange market.** The foreign exchange market will be covered in the chapter on international trade and finance.

The money market

The money market is the financial market segment that deals in very short term overnight to six months funds. Banking institutions are the key players in the money market. The market segments of the money marketing Kenya include the interbank market, the government securities market and private securities like commercial paper, certificate of deposit and bills of exchange. Money market participants in Kenya include the government, the central bank, the supervisory authority, 50 commercial banks, 21 non-bank financial institutions, building societies, insurance companies and corporate entities. The role of the money market in the economy is as follows:

1. It mobilises short-term funds to support economic activities by the government and the private sector.

2. It is a medium for the central bank monetary policy such as managing the banking system liquid influencing short term interest rates, controlling inflation and influencing developments in the exchange rate.

3. The money market provides a medium for commercial banks to manage their funds by allowing them to lend temporary surplus funds and borrow funds to support their lending commitments.

The role of financial intermediaries

Financial intermediaries are organisation, which channel funds from institutions and individuals who have financial surpluses to institutions and individuals which wish to borrow funds. They therefore intermediate between lenders and borrowers, earning their profits from the difference in the rate of interest paid to depositors and the rate of interest charged to borrowers. The main financial intermediaries in the Kenya economy are commercial banks, insurance companies, and building societies, unit trusts, pension funds, and others. **Disintermediation** describes the bypassing of financial intermediaries so that borrowers and lenders deal with each other with the possibility that costs can be saved.

The functions of commercial banks

The following are the **functions** of commercial banks in an economy

1. Commercial banks provide a payment mechanism through which individuals, firms and government can make payments to each other. Modern banks utilise a payments clearing system which enable s individuals and firms to make payments by cheque. Commercial banks also constitute a source from which individuals and firms can obtain notes and coins.

2. Commercial banks also provide a place for individuals, firms and government to sore their wealth, for example, in the form of current or deposit accounts. Commercial banks also compete in order to attract funds from different individuals and organizations.

3. Commercial banks lend money in the form of loans or over generate a substantial proportion of heir incomes.

4. Commercial banks act as financial intermediaries by accepting deposits and lending to individuals and organisations that require finances.

5. Commercial banks also act as foreign exchange dealers. They provide a means of obtaining foreign currencies or selling foreign currencies. They make their profits on the basis of the spread, which is the difference between the buying, and the selling price of the foreign currencies.

6. Commercial banks may also provide their clients with expert advice on a broad rnge of matters such as those relating to investment, takeover bids, share registration and leasing.

The role of non-bank commercial intermediaries

Non-bank financial intermediaries operate in both money and capital markets depending upon the type of financial institution. The rationale for their participation in money markets is similar to that of commercial banks. However, these institutions tend to concentrate their borrowing and investment in instruments different from those used by commercial banks. For example, mutual savings bank, life insurance companies and building societies are suppliers of long term funds in the capital markets:primarily when they purchase and sell stocks and bonds. Finance companies, especially the deposit taking types, are short-term demanders and suppliers, of funds, often borrowing on short-term basis from banks to make short-term consumer loans. They act as intermediaries between deficit and surplus units of the economy in the financial intermediation process which is the essence of the role of financial institutions within the markets and the general economy at large.

The **role of non-bank financial intermediaries** in the Kenyan economy is as follows:

1. They stimulate competition with commercial banks over deposits and over the credit market, which stimulates efficiency in terms of improved services for savers and borrowers in the financial market.

2. They have enhanced the development of the financial market through the introduction of a great variety of financial instruments. Thus, for example, deposits such as investment deposits accounts, mortgage deposit accounts, savings and credit union members' contribution accounts, building societies member' accounts feature in the financial market. A greater variety of financial instruments is desirable for the development of financial markets in order to cater for differences in savers' and borrower's' interests.

3. Commercial banks traditionally lend out funds on a short-term basis to safe borrowers and thus do not effectively cater for long term, risky borower market whereas non-bank financial intermediaries often lend out on longer-term basis and to risky areas, which explains why, they often charge high rates of interests than commercial banks.

4. Non-bank financial intermediaries often provide financial services, which are beyond the scope of commercial banks such as chattel mortgage loans and purchase finance.

5. The development of non-bank financial intermediaries through its expansion of the financial markets has created an additional vehicle for the more effective execution of the government's monetary policy.

The major difference between commercial banks and non-bank fiancial institutions are as follows:

1. Commercial banks operates cheque accounts, which make them members of the central bank clearing houses whereas non-bank financial institutions are not members of the central clearing houses since they do not operate cheque accounts.

2. Commercial banks can create credit by allowing cheques to be drawn on them in excess of the amounts deposited with them that have bee deposited with them.

3. Commercial banks usually accept short term deposits and lend on short term, relatively secure whereas non bank financial intermediaries ma accept long term

deposits and lend on relatively long term and more risky ventures which enables them to charge higher interest rates.

4. Commercial banks operate bank accounts with the central bank, which is the banker's banks whereas non-bank intermediaries do not have this facility.

5. Some deposits placed with commercial banks, notably current account deposits, are no-interest bearing whereas all deposits placed with non-bank financial intermediaries are interest bearing.

INTERNATIONAL TRADE

International trade is the exchange of goods and services between one country and another. The exchange of goods and services across international boundaries has enabled the principle of division of labour to be extended to the international sphere.

International trade arises because:

1. The production of different kinds of goods requires different kinds of resources used in different proportions

2. The various types of economic resources are unevenly distributed throughout the world.

3. The international mobility of resources is extremely limited.

Since it is very difficult to move resources between nations, the goods, which 'embody' the resources, must move. Just as our individual abilities and aptitudes fit us for different occupations, so the different resources of historical development of nations equip them for the production of different products. Unlike individuals, nations do not specialise completely in one procedure or in one product. But they tend to concentrate on certain types of activity.

The following are the basic **advantages** of international trade"

1. International trade leads to an increase in the standard of living of all people in the countries themselves. It enables consumers in all countries to have a larger quantity of goods and services than they would have had if they tried to be self-sufficient and specialisation itself increases output.

2. Some countries are able to import machinery and other factor inputs necessary for industrialization. Exports, therefore, generate valuable foreign exchange for economic development.

3. International trade also encourages efficiency since all producers involved in international trade will try to be more efficient in order to be competitive both in domestic and foreign markets.

4. International trade will also promote peace and security.   This is because countries are mutually interdependent and d so they will try to avoid the outbreak of war.

5. International trade will also enlarge the commodity market for a country's commodities of scale due to the advantage of a large market.   Economies of scale will contribute to fuller utilization of capacity and new industries may be set up which were not possible before because of a limited domestic market.

6. International trade broadens the consumer's choice, providing them with access to a wider range of commodities from abroad. This greater variety increases the utility of consumers.

The concept of absolute advantage

A country is said to have an absolute advantage in the production of a particular commodity if with a given quantity of resources it can produce ore of that commodity than can any other country using the same quantity of the same resources. Thus if country A can produce more of a commodity with the same amount of real resources than country B(this implies at a lower absolute unit cost) country A is said to have an absolute advantage over country B.

Let us assume for instance that it takes 100ma n-hours to produce 1 ton of rice in China and 300 man-hours to produce 1 ton of rice in Kenya. This means that 100 man-hours in Kenya will produce only one third ton of rice. Since a given quantity of labour (100 man-hours) produces more rice in China than in Kenya. China is said to have an absolute advantage over Kenya in the area of rice production.

If country A has an absolute advantage over another country B in 1 commodity while B has an absolute advantage over A in another commodity, the total production of both can be increased by each country specialising in the production f the commodity in which is has an absolute advantage. It is important to note that trade was necessary to achieve the gains from specialization.

The concept of absolute advantage can be illustrated in the example that follows:

It is assumed that there are two countries namely country 1 and 2 and two commodities, A and B.

Country	**Cost of producing one unit (in man-hours)**	
	A	B
1	5	10
2	10	5

In the example above, Country 1 has an absolute advantage over Country 2 in the production of commodity A since it can produce unit of commodity A with 5 man-hours, whereas country 2 can only produce one-half unit of commodity A with the same resources (man-hours). On the other hand country 2 has an absolute advanced in the production of commodity B since it can produce 1 unit commodity B with 5 man-hours, whereas country 1 can only produce one-half unit of commodity B with the same level of resources (man-hours).

The concept of comparative advantage

A country has a comparative advantage over another if in producing a commodity it can do so at a relatively lower **opportunity cost** in terms of forgone alternative commodities that could be produced. Taking 2 countries, A and B, each producing 2 commodities, X and Y, country A is said to have a comparative advantage in the production of X if it can produce commodity X at a relatively lower opportunity cost than country B. this implies that its absolute margin is greater or its absolute disadvantage is less in commodity X than Y.

Try A is therefore efficient country, it has an absolute advantage over country B in both industries, it can be shown that even under these circumstances it is possible for specialization and trade to benefit both countries. The principle involved is known as the **principle comparative costs.** This states that even where one country has an absolute advanced over the other in both industries, specialization and trade can still benefit booth countries provided each country has a comparative cost advantage. Comparative cost relates to the opportunity costs producing the commodities and not the absolute costs. The concept of comparative advantage can be illustrated in the example that follows:

Country	**Cost of producing one unit (in man-hours)**	
	A	B
1	8	9
2	12	10

The opportunity cost of good A is the amount of other goods, which have to be given up in order to produce one unit of the good. To produce a unit of good A in country 1, you need 8 man-hours and 9 to produce goo B than A. The opportunity cost of producing good A is equivalent to 8/9 units of B. One unit of B is equal to 8/9 units of A in country 2, one unit of A is equal to 12/10of B and one unit of B=10/12 units of A.

Country	**Cost of producing one unit of**	
	A	B
1	8/9(0.89)B	9/8(1.25)A
2	12/10(1.2)B	10/12(0.83)A

B is cheaper to produce in country 2 in terms of resources as opposed to producing it in country1. A is cheaper to produce in country 1 than in country 2 in terms of how much of commodity b is given up.

Therefore, the following conclusions can be made:

1. A country has a comparative advantage in producing a commodity if its opportunity costs are lower.

2. Opportunity costs depend on relative costs of producing 2 commodities and no on absolute costs.

3. If opportunity costs are the same in both countries, there is no comparative advantage and therefore, no possibility of gains from trade and specialisation

4. If opportunity costs differ in the tow countries and both countries are producing both commodities, it is always possible to increase the production of both commodities by a suitable reallocation of resources within each country.

The fundamental generalisation of the theory of gains from trade can now be stated as:whenever an opportunity costs differ among countries, specialisation of each country by producing those commodities in which it has a comparative advantage will make it possible to increase production of all commodities relative to the quantities available to them if each country attempted to be self-sufficient. The principle of comparative advantage recommends that a country should specialise in the production of only those commodities in which it has a comparative advantage and export surpluses of such commodities to the rest of the world.

UNEMPLOYMENT AND THE UNEMPLOYMENT RATE

Unemployment refers to a situation where factors of production are willing and capable of being employed at the ruling market wage rates but are involuntarily unutilized or under-utilized.

The unemployment rate is increased by, for example, lay-off and a greater number of school leavers without a job. It is decreased by unemployed people finding jobs and laid off workers being employed.

The unemployment rate only tires to measure unemployment that is involuntary. The number of unemployed individuals will depend on the rate of expansion of work opportunities which in turn depends on the rate of expansion of work opportunities which in urn depends on the rate of growth of the economy, the production technique utilised and government policies. One of the difficulties of trying to measure the unemployment rate in developing country is the considerable size of informal sector where activities are rarely portrayed accurately by employment statistics.

Types of unemployment

1. Open involuntary unemployment

This occurs when a person is willing to work at the ruling wage rate but is not able to secure a job. This concept is particularly relevant in the modern urban sector where young people aspire to obtain white-collar jobs and are unable to do so.

2. **disguised or 'hidden' unemployment**

occurs when the work available to a given workforce is insufficient to keep it fully employed so that some members of the workforce could e withdrawn without loss of output. An example f this type of unemployment relates to the civil service in many developing countries where the number of people hired often exceeds those that are required. The marginal product of labour in these cases is zero and does not contribute to national output, although such unemployment is not recorded in official statistics. Disguised unemployment is also common in rural areas in developing countries where agriculture is practiced. May such individuals working in small plots of land are in fact in disguised employment since they could be withdrawn without a fall in output because their marginal product is zero or even negative.

3. **General unemployment** is that which is widespread throughout the economy and not confined to particular regions of categories of labour.

4. **Structural unemployment.** Unlike general unemployment, is that whch affects particula regions or categories of labour and results from an imbalance between the supply for a particular group of workers and the demand for their services. An example of how such an imbalance would occur is where technological change makes the product on which a particular industry is based obsolete or alternatively, new methods of production render labour with particular skills redundant. On the demand

side, changes in consumer tastes, compensation from substitute products or new products in different areas may be responsible.

5. **Seasonal unemployment** Regular seasonal unemployment is caused by annual variations of seasons, which affect economic activities in sectors such as in agriculture, fisheries, construction and tourism. During the pick seasons in these sectors the demand for labour will be very high whereas during the off-peak or low seasons there will be standard drop in this demand.

6. **Frictional unemployment** This is unemployment which arises from immobilities in labour force rather than from a lack of demand for labour. It is essentially short term in nature and includes unemployment, which arises when people are changing jobs or because of a lack of knowledge about job opportunities. It usually takes time to match prospective employees with employers and individuals will be unemployed during the search period for a job.

7. **Demand-deficient or cyclical unemployment.** This type of unemployment is associated with the trade cycle. During the recovery and boom phases of he trade cycle the demand for output and labour is high and unemployment is low. On the other hand, during the recession and depuration years, the demand for output and labour falls and unemployment falls rises sharply,. Demand –deficient unemployment can be relatively long-term in nature. Keynes directed his attention to this type of unemployment, which he believed could be eradicated by demand management policies.

Causes of unemployment in developing countries

The following are the main causes of unemployment in developing countries:

1. **Lack of co-operating factors of production.** These are other essential factors, which are combined with labour in production process. A firm can only employ additional labour if it has more of cooperating factors like capital and land. Developing countries, however, lack these factors especially capital and foreign exchange.

2. **Rapid population growth** the problem is that in many developing countries, population grows faster than the overall rate of growth of the economy. This implies that the labour force entering the labour market expands faster than the absorption capacity of the economy.

3. **The use of inappropriate technology.** Technology is said to be inappropriate in relation to the resource based in a given country. In developing countries, technology is often labor saving or capital-intensive which is inappropriate sine most developing countries are labour surplus economies. A high capital-labour ratio implies less labour is absorbed in production processes thereby causing unemployment.

4. **Distortion of relative factor prices.** Following the principle of economy, producers are assumed to face a given set of relative factor prices and to utilize the combination of labour and capital that minimizes the cost of producing a given level of output. In

many developing countries governments distort prices making capital relatively cheaper than labour through, for example, various public policies such as investment incentives, tax allowances, subsidized interest rates and low tariffs on capital imports. On the other hand, labour is made artificially expensive through minimum wage laws and trade unions.

5. **The nature of the education system.** Education systems in many developing countries were adopted from developed countries and are geared to white collar jobs, which do not conform to he realities of the labour market. The rate of job creation in the formal sector in developing countries where white-collar jobs are found is much lower than the number of people entering the labour force. Consequently white-collar jobs are increasingly hard to come by.

6. **Seasonality in production** this factor is especially important in developing countries where the agricultural sector is predominant. Changes in weather lead to seasonality in agricultural production and hence seasonality in employment. Seasonal unemployment is also prevalent in the tourism sector where tourists prefer to travel during certain times of the year.

7. **Massive rural to urban migration** resulting in urban unemployment owing to insufficient urban job creation capacity. Rural to urban migration is the physical movement of people from rural to urban centres of a country. High rural to urban migration can be explained by the following factors:

- The perception of the existence of income differentials between the two areas. This explains the standard"push"from subsistence agriculture and "pull" of relatively high urban wages.

- The migrants perceive high chances of getting jobs in the urban centres partly because of the concentration of industrial production in urban centres.

- The non-availability of social amenities in rural areas

- Pressure on limited land in rural areas.

- Social factors including the desire of migrants to break away from the traditional constraints of social organisations

- Physical factors including climate and meteorological disaster like drought and floods which tend to be more prevalent in the rural areas.

- Demographic factors including a fall in mortality rates and consequent high rates of rural population growth.

- Communication factors such as improved transportation urban-oriented education systems and the "modernising" effect of the introduction of radio, television and cinema.

The adverse effects of economic reform programmes. May developing countries are undertaking economic reform programmes which entail the liberalisation of certain key sector, for example, textiles. Such liberalisation may contribute to domestic unemployment if domestically produced commodities are for example, unable to compete with cheaper foreign substitutes. This is because some domestically owned firms may be forced to shut down their operations and consequently lay off may workers. In addition, economic before programmes may have a component of Civil Sector Reform which may lead to the retrenchment of may workers in the civil service with consequent rising unemployment.

Reasons for concern about unemployment

1. Unemployment represents a waste of potentially productive human resources. When labor is unemployed this implies that the economy is not producing as much as it could. This in turn signifies that national income is lower than it could be and, therefore, national welfare is lower where unemployment is high.

2. Greater unemployment means a higher dependency ration since the few employed have to support a large number of dependants.

3. An increase in social problems like crime. Unemployment also contributes to the social problems of personal suffering and distress which can lead to a rising incident of mental disorders.

4. Overcrowding in urban areas resulting from urban unemployment. This is prevalent in urban areas with high rates of rural-urban migration.

5. Unemployment also represents a loss of human capital since the unemployed labour will gradually lose its skills. This is because skills can only be maintained by constant work and practice.

6. In order to maintain the unemployed, the government may be forced to increase its expenditure on social amenities. This constitutes a drain on its expenditure which could be used for other development projects.

Policies to combat unemployment in developing countries.

Government policies to influence employment may either aim to reduce the total number of unemployed people or to increase the level of job creation. Policies to reduce unemployment are linked to the causes of unemployment. These policies may take the following forms of supply side and demand side policies.

1. Increasing employment creation in the private sector

Although the public sector in many developing countries is the largest employer, the potential for employment creation for this sector is limited. Government policy can support private sector development by creating an enabling environment for private sector development. Reduced budge deficits will have the effect of lowering interests rates and facilitating private sector investments. The government can also provide incentives to microfinance institutions which provide capital to small and medium enterprises. Of particular importance in employment creation is the informal (jua kali) sector given that is limited. The availability of cheap finance can help firms to acquire co-operating factors of production and hence reduce levels of unemployment.

2. Pricing policies that encourage the use of appropriate technology

Pricing policies can play a leading role in guiding investment towards labour-intensive technologies in different sectors, which are appropriate to the labour surplus economies in many developing countries. Government policies should aim at lowering the relative price of labour to create greater employment. The impact of removing factor price distortions on employment creation will depend on the degree of sustainability of labour for capital in the production processes of various sectors of the economy.

3. relevant education systems be adopted

The education systems adopted in developing counties should emphsise skills required by the labor market. Indiscriminate education expansion contributes to the unemployment. Education expansion should, for example, balance the need to provide general education and professional skills since the latter are often more readily marketable.

4. seasonal employment

the diversification of economic activities can be instrumental in reducing seasonal unemployment. For example, regions that depend on tourism for employment can introduce alternative activities such as labor-intensive manufacturing during the off-peak season. Seasonal unemployment in agriculture can also be reduced by engaging in industrial activities, where possible.

5**. Diversification of products and markets**

unemployment resulting from limited product markets can be reduced by diversifying from primary products into other lines of production where demand is more price and income elastic. This could partly include further processing of primary products to add value to them. However, such diversification should still utilise labour-intensive technology for substantial employment creation to take place. Firms should also seek new markets for their products if current markets are unlikely to expand substantially. Since the demand for labour is a derived demand, it will increase if the demand for the commodities produced by labour increases.

5. Intensive rural development

The long-term way to combat rural-urban migrations is by emphasising rural development. This can be done by the government providing incentives for industries to locate in the rural areas thereby increasing employment and standards of living in rural areas. Social amenities in rural areas should also be improved. In this way, the incentives to migrate to urban areas will be reduced with a consequent decrease in both urban and rural unemployment.

6. encouraging foreign direct investment

Developing countries should aim to make the political and economic environment conducive to the inflow of foreign capital. This is because such foreign capital can contribute considerably to enhance domestic employment opportunities. This is especially so where the foreign firms use technology which tends to create greater employment.

7. Encouraging the use of domestic resources.

Developing countries should promote domestic resource use whenever possible because this tends to create employment domestically. Considerable use of foreign inputs should be reduced, where possible, since such usage generates employment abroad.

THE ROLE OF AGRICULTURE IN ECONOMIC DEVELOPMENT

- Agriculture contributes a large proportion of the gross product in many developing countries and it is therefore relatively easier to achieve a higher rate of economic growth by expanding agriculture. Thus, for example, if agriculture contributes 75% of Gross Domestic Product (GPD) and industry constitutes 25% of GDP then it is easier to achieve a target rate of economic growth by expanding agriculture.

- Agriculture is a more appropriate activity to develop since most developing countries are labour surplus economies. Agriculture, therefore, facilitates a fuller utilisation of the resource base and provides greater employment opportunities.

- Increasing agricultural output and income is often a prerequisite for the expansion of the industrial sector since as worker in the agricultural sector earn higher incomes, particularly small holder producers, their demand for manufactured goods will increase and there will be greater scope for establishing industries. This implies that the rate of industrialization will to some extent depend on how rapidly agricultural incomes are increasing. Continuing economic development in urban areas will considerably restrict the domestic market for manufactured goods. Increasing the purchasing power of rural areas as a result of expanded agricultural output will raise the demand for manufactured goods. Increasing the purchasing power of rural areas as a result of expanded agricultural output will raise the demand for manufactured goods.

- An increase in agricultural incomes can boost government revenues especially where the agricultural sector forms a broad base for taxation. His revenue could enhance economic, social and infrastructural development.

- A number of industries are based on agricultural raw materials such that an expansion of industries relies on more raw material being produce. The textile industry, for example depends on cotton and wool production. The development of domestic agriculture would, in addition, reduce domestic reliance on imported raw materials.

- Agricultural development is an important step towards self-sufficiency in food production, which is important in keeping a growing population alive. In addition, this would save on valuable foreign exchange being used on food imports. The increase in agricultural production should be at a higher rate than the rate of increase in food demand in order to avoid increases in food prices ad the need to import food from abroad.

- Increased agricultural production will stimulate forward linkages with other sectors of the economy through demand for transport, services, marketing and processing as well as some backward linkages in the form of farm inputs and equipment.

- Agricultural products often constitute significant exports in developing countries. Agriculture therefore provides the needed foreign exchange to import essential inputs especially of capital goods. These capital goods facilitate the industrialization process and tend to e in short supply in developing counties. As a country industrialises, the proportion of agricultural exports in its total exports is likely to fall.

- The agricultural sector also contributes an important source of savings. Since in many developing countries a considerable proportion national income is derived from agriculture, in the short run at least, the savings potential will be in the agricultural sector. Savings are vital for economic growth in that they are a source of vital investment funds.

- The agricultural sector also provides productive employment in many developing countries. This is especially so here the agricultural production technology has remained fundamentally labour-intensive. Increased emphasis has also been placed on employment creation in the agricultural sector given the high rate of rural-urban migration and the inability of urban-based economics sectors to create sufficient employment.

Problems faced by the agricultural sector in developing countries

- Agricultural products are subject to frequent price fluctuations which in turn lead to fluctuation in farmer's incomes.

- Agricultural products are subject to long run declining terms of trade compared to manufactured goods. The impact o the factor is worsened since developing countries mainly export agricultural products and import highly priced manufactured goods.

- Developing countries usually depend on the export of one or two main agricultural products, which make their economies particularly vulnerable to changes in international economics conditions. For example, if a country is heavily dependent on coffee exports and world coffee market prices decline sharply that country's export revenue will also fall substantially.

- The agricultural sector is particularly subject to the law of the agricultural sector is particularly subject to the law of diminishing returns especially in developing countries where population pressure on land is high leading to disguised unemployment.

- Agricultural products from developing countries are faced by protectionist barriers such as tariffs from developed country markets. Developed countries seek to achieve self-sufficiency in food production and to protect the employment of their farmers. Such protectionist measure limit the growth of agricultural exports in developing countries and the potential that such benefits would provide.

- Many farmers in developing countries still use inefficient traditional methods of production such as land fragmentation, which severely limits increase in productivity in the agricultural sector.

- Marketing channels for agricultural products in many developing countries are inefficient and inadequate. May products are marketed by marketing boards, which are often plagued by managerial problems, corruption and inadequate finances to effectively exercise their functions.

- Many farmer sin developing countries are subsistence farmers and lack the necessary capital and resources to effectively implement modern agricultural development such as the use of mechanised agriculture.

- The agricultural sector is particularly vulnerable to natural disasters such as floods, droughts, diseases and pests, and this leads to wide annual fluctuations in output.

- Agricultural production is seasonal by nature and hence the agricultural sector is characterised by seasonal unemployment. The seasonal nature of agricultural employment contributes t many individuals seeking jobs in alternative sectors that provide relatively stable incomes.

- The discovery of synthetic products has contributed to some extent to the replacement of some agricultural commodities such as sisal. This has in turn reduced the demand for these agricultural commodities.

Policies to improve agricultural sector

- Use of buffer stocks and buffer funds where necessary in order to stabilise agricultural prices and incomes. These policies were covered in Chapter Three dealing with elasticities.

- To deal with declining terms of trade developing countries should diversify their economies by improving their industrial base and move into non-traditional lines in agriculture, such as, horticulture. In Kenya, for example, successful diversification has been achieved into horticulture. A greater diversification into industry has also been achieved in Kenya with manufacturing exports becoming more important in terms of their contribution to the export sector. Developing countries can also diversify their economies by further processing of commodities instead of simply exporting raw materials.

- The use of more efficient methods of production should be encouraged by educating farmers through extension services. Extension services will ensure that the results of agricultural research reach many farmers. Extension workers can, therefore, be important change agents. In order to be successful extension services should avoid being excessively bureaucratic and should also focus on providing favourable conditions such as prices to encourage farmers to adopt technologies that have been demonstrated. Extension worker should both transmit knowledge and promote the adoption of new technology.

- Credit facilities should be provided to farmers in terms of soft loans to enable them to purchase needed capital inputs, microfinance institutions can play a valuable role in this regard in developing countries, as can co-operatives.

- A more liberalised system should be introduced where the private sector plays a role in production, marketing and processing. This will help to overcome the inefficiencies that have been associated with marketing boards in developing countries.

- Research facilities should be developed to increase crop output and improve crop quality. Agricultural research is a major factor that contributes to greater agricultural productivity. Such research that contributes to greater agricultural productivity. Such research can focus on aspects such as irrigation practices, crop rotation and optimal planting times. Domestic agricultural research is vital in adopting imported technology to local conditions.

- Where funds are available, marginal lands should be irrigated in order to reduce pressure on fertile land and to delay e operation of law of diminishing returns.

- Markets should be diversified to non-traditional markets and by regional integration efforts. Kenya exports have, for example, been concentrated in the markets of the European Union and the East African region. Economic integration effort such as COMESA and the East African Community provide prospects for more diversified exports markets.

Summary of agriculture contribution to economic development

1. Agriculture provides more food to the rapidly growing population.
2. Increases the demand for industrial products and thus necessitating the expansion of the secondary and tertiary sectors
3. Provides additional forex earnings for the import of capital goods for development through increased agricultural exports.
4. Increases rural incomes to be mobilized by the state.
5. Provides productive employment
6. Improves the welfare of the rural people – peasants consume more food of higher nutritional value, build better homes etc.
7. Demand for inputs like fertilizers, better tools, implements, tractors, irrigational facilities in the agricultural sector will lead to greater expansion of the industrial sector.
8. Means of transport and communication will expand when agricultural surplus is to be transported to urban areas and manufactured goods to rural areas.
9. The expansion of secondary ad tertiary sectors due to increased productivity in agriculture leads to the increase in the rate of capital accumulation.
10. As development gains momentum due to industrialization, the proportion of agricultural exports in a country’s total exports is likely to fall as they are needed

in larger quantities for domestic production of imported articles. Such articles are import substitutes and conserve forex. Larger production of food crops and export crops conserves and earns forex and lead to expansion of other sectors of the economy.

11. Forex earnings can be used to build efficiency of other industries and help the establishment of new industries by importing scarce raw material, machines, capital equipment and technical knowhow.
12. Due to increased productivity in agriculture, the surplus labor can be used for construction works.
13. Due to increased agricultural productivity, the government can mobilize increased farm incomes through agricultural taxation, land taxes, agricultural income tax, land registration charges, fee for providing agricultural technical services etc.
14. Agricultural expands and diversifies employment opportunities in rural areas. As agricultural productivity and farm income increase, non-farm rural employment expands and diversifies. Landless and marginal farmers are primarily engaged in non-agricultural pursuits which include the manufacture of textiles, furniture, tools, handicrafts, leather and metal working, processing, marketing, education, construction of houses and other buildings etc.

Rural people consume more food of a higher nutrition value, i.e., meat, eggs, milk etc. They build better homes with modern amenities like electricity, furniture, radio etc. They provide themselves with bicycles, motor cycles, ready made garments shoes etc. They receive direct satisfaction from such services as schools, health centers, irrigation, banking, transport and communication etc.